

On Prediction Markets as Instruments of Market Completion

Anna Orlik and Cristina Scofield*

June 3, 2026

Abstract

Prediction markets are conventionally framed as information aggregators—mechanisms that elicit dispersed beliefs through trading. We argue this framing obscures a distinct welfare rationale: binary event contracts function as Arrow securities that expand the span of state-contingent claims, enabling risk-sharing even when no private information is aggregated. We develop this *market completion* rationale in a two-period Arrow–Debreu framework and quantify the magnitude of welfare gains in two calibration exercises. The welfare case for Kalshi-style contracts does not primarily rest on total inflation-risk volume—that is well-hedged by existing assets—but on the presence of agents with *non-linear* exposure that smooth linear hedges cannot span. A two-agent toy calibration establishes that completion gains can reach several percentage points of certainty-equivalent consumption; a scaled-up, empirically disciplined eight-agent calibration shows that once TIPS are in the baseline asset menu these gains shrink to roughly 0.003–0.05% of aggregate certainty-equivalent consumption at plausible shares of threshold-exposed agents, because linear inflation risk is already well hedged and only non-linear (threshold-crossing) exposures remain unspanned. We then bring the completion logic to data through asset-pricing spanning tests on a panel of Kalshi macroeconomic contracts: regressing daily contract price changes on a broad menu of investable hedges—inflation-protected, nominal-bond, and equity ETFs, ten-year nominal and breakeven yields, and constant-maturity Fed funds futures—we find that roughly 95% of daily Kalshi price variance is left unexplained, even for the most hedgeable family augmented with its most direct tradeable instrument. This spanning failure satisfies the necessary orthogonal-complement condition that the completion rationale requires.

1 Introduction

Prediction markets have rapidly expanded in popularity and trading volume in recent years, transitioning from an academic curiosity to a regulated financial product. These markets offer exchange-traded binary contracts that pay a fixed amount (typically \$1) if and only if a specified event occurs. As seen in Figure 1, Kalshi, the first federally regulated prediction market in the United States¹,

*Board of Governors of the Federal Reserve System (corresponding author’s email: anna.a.orlik@frb.gov). We thank seminar participants at the Board of Governors of the Federal Reserve System. The views expressed in this paper are solely the responsibility of the authors and should not be interpreted as reflecting the views of the Board of Governors of the Federal Reserve System or any other person associated with the Federal Reserve System. Parts of this paper were completed using AI models.

¹Prediction markets writ large are not necessarily a new phenomenon. Several smaller-scale markets already operated within the US through “no action” letters from the CFTC such as the Iowa Electronic Markets and PredictIt, both of which operated with substantially smaller investment limits than we see in larger markets like Kalshi and Polymarket.

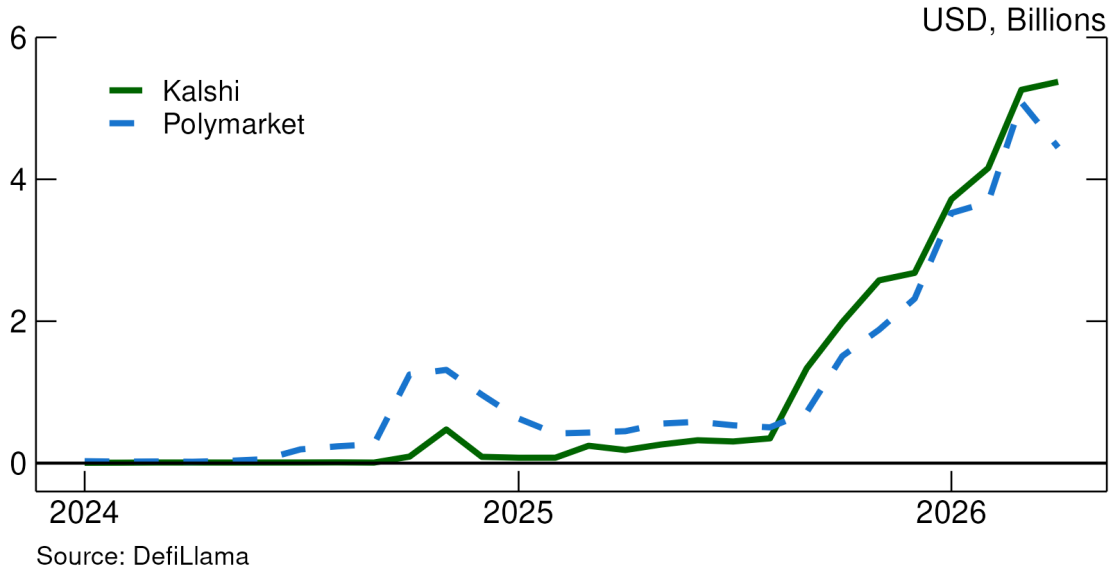


Figure 1: Market Volumes, January 2024 - April 2026

grew from roughly \$300 million in 2024 cumulative notional volume to approximately \$15 billion in monthly notional volume by mid-2026. Polymarket, the leading blockchain-based venue, cleared over \$2.5 billion in trades in the three months surrounding the 2024 U.S. presidential election alone, as seen in Figure 1.²

A cascade of U.S. regulatory and judicial decisions—Kalshi’s landmark November 2020 designation as a Designated Contract Market, the October 2024 U.S. Court of Appeals for the District of Columbia ruling in *KalshiEX LLC v. CFTC* vacating the CFTC’s prohibition on congressional-control event contracts, the CFTC’s May 2025 voluntary withdrawal of its appeal, and the July 2025 designation of Polymarket US as a Designated Contract Market—have substantially liberalized the categories of events for which US exchange-listed contracts may be written. With this expansion has come renewed academic attention to the trading dynamics within prediction markets [Gomez Cram et al., 2026], accuracy of their forecasts [Bürge et al., 2026], and a growing empirical literature using Kalshi and Polymarket data to study macroeconomic expectations [Diercks et al., 2026, Eichengreen et al., 2025, Swanson et al., 2025].

Yet despite this renewed attention, the intellectual case for prediction markets remains relatively one-dimensional. The dominant framing, visible in both popular treatments [Wolfers and Zitzewitz, 2004] and policy advocacy [Arrow et al., 2008], casts prediction markets as *information aggregators*: mechanisms through which trading elicits dispersed private beliefs and compresses them into a single, publicly observable price. This framing runs from Hayek [1945] through Grossman [1976] and Grossman and Stiglitz [1980], and the core empirical claim is that prediction market prices

²We reference cumulative volume and show volume data for Polymarket (combined international and US exchanges) and Kalshi from defillama.com

forecast well, and often better than polls or expert judgment.

This paper argues that a second, logically distinct, and underdeveloped rationale deserves equal attention. Prediction market contracts are, in their most basic form, Arrow securities: contracts paying one unit of numeraire conditional on the realization of a particular state. The classical welfare theorems of [Arrow \[1964\]](#) and [Debreu \[1959\]](#) establish that a full set of such securities is sufficient for Pareto efficiency under standard conditions. Real-world financial markets are conspicuously incomplete: consequential sources of risk—election outcomes, central bank decisions, regulatory approvals, pandemic declarations, macroeconomic threshold crossings, and extreme weather events—cannot be directly hedged through equities, bonds, or standard derivatives. Prediction markets (in specific forms that will be discussed shortly) have the potential to fill the gaps this framework identifies as missing. Their welfare contribution is not, or not only, the production of information, but also the provision of risk-sharing instruments against state-contingent exposures that currently go unhedged.

The distinction matters for three reasons. First, the two rationales rest on different modeling primitives and can move in different directions: [Hirshleifer \[1971\]](#) showed that information production can disincentivize participation in insurance markets, so the activities of aggregation and completion can be substitutes rather than complements. Second, the regulatory case for prediction markets—which typically depends on whether contracts have “economic purpose” under the Commodity Exchange Act—is stronger under the completion framing, because hedging demand is economic purpose in the narrow statutory sense. Third, the potential empirical program differs: forecasting accuracy has been exhaustively studied, but the welfare gains from completion remain almost entirely unmeasured. Indeed, recent empirical work has cast some doubt on the aggregation story: [Clinton and Huang \[2025\]](#), studying 2,500 political contracts across Kalshi, PredictIt, and Polymarket during the final weeks of the 2024 U.S. election, find substantial price divergence across venues, evidence of herding on high-volume platforms, and arbitrage opportunities that peaked in the final two weeks before Election Day—findings that raise questions about the efficiency of aggregation but do not bear directly on the completion function, which is the subject of this paper.

That said, there is a wide range of contracts and topics offered on popular prediction market exchanges; alongside financial, economic, and political outcomes, users can also purchase contracts predicated on the outcomes of sporting events, Oscar award contests, and the timing of celebrity weddings. While there is significant volume in these categories, their case as risk-sharing instruments is less tractable than for contracts that are more directly economically relevant. Further, making a welfare case for these types of less economically relevant contracts also involves taking a stand on whether the welfare benefits obtained by the availability of these contracts outweighs the potential negative consequences of making available what some argue is simply widespread, easily accessible, and relatively unregulated gambling. For the purposes of this paper, we henceforth restrict our discussion of prediction market event contracts to the subset that are directly economically relevant, in this case inflation, unemployment and other related macroeconomic and financial contracts, and abstracting from relevant discussions of insider trading or market manipulation.

The remainder of the paper proceeds as follows. Section 2 reviews the two traditions separately and documents their conflation in the applied prediction-markets literature, integrating recent empirical work on Kalshi and Polymarket. Section 3 develops the completion framework, delineates separation from aggregation, and engages with the negative results on financial innovation in incomplete markets. Section 4 presents a two-agent toy calibration that establishes a quantitative lower bound on welfare gains from a Kalshi-style inflation contract. Section 5 extends the calibration to an eight-agent economy with empirically disciplined heterogeneous inflation exposures and documents a refined finding: Kalshi’s marginal contribution on top of TIPS rests specifically on agents with non-linear exposures. Section 6 brings the completion logic to data through asset-pricing spanning tests on a panel of Kalshi macroeconomic contracts, establishing that conventional hedges leave the bulk of Kalshi price variance unspanned. Section 7 concludes and outlines directions for future empirical work. Appendix A provides a rigorous proof of Proposition 1.

2 Literature Review

2.1 The Information-Aggregation Tradition

The information-aggregation view of markets traces to Hayek [1945], who argued that the price system’s central function is the coordination of dispersed knowledge. Grossman [1976] formalized the idea in a rational-expectations equilibrium where prices reveal aggregated private signals, and Grossman and Stiglitz [1980] identified the informational free-rider problem that prevents full revelation when information acquisition is costly.

Within this literature, prediction markets occupy a distinct position. Because their contracts pay out based on objectively verifiable public events, the mapping from true state to asset payoff is unambiguous, and the noise-trading, liquidity, and heterogeneous-beliefs complications that plague asset-market interpretation are muted. Wolfers and Zitzewitz [2004] survey the empirical forecasting performance of prediction markets across political, economic, and sports events and document consistent accuracy relative to polls, pundits, and statistical benchmarks. Berg et al. [2008] and related work on the Iowa Electronic Markets extend this documentation to election forecasting specifically. Manski [2006] offers an important caveat: observed prices in prediction markets correspond to mean beliefs only under strong assumptions on trader preferences and beliefs, and more generally bound the underlying distribution rather than pinpoint it. Ottaviani and Sørensen [2015] examine systematic biases, including the favorite-longshot bias, that complicate the interpretation of prices as probabilities.³

An extension of this literature concerns mechanism design for thin markets. Hanson [2003, 2007] proposes market scoring rules as a hybrid of proper scoring rules and market-making that retains aggregation properties even when simultaneous participation is low. This body of work is technically elegant but addresses the information aggregation question exclusively: it asks how

³Bürgi et al. [2026] document the favorite–longshot bias within Kalshi markets as well.

to extract information when natural trading would fail, not whether the resulting contract set is welfare-improving for reasons independent of information.

2.2 Recent Empirical Evidence on Kalshi and Polymarket

The 2024–2026 period has produced the first substantial body of transaction-level empirical work on prediction markets, most of which remains focused on information aggregation and forecasting accuracy rather than welfare. We summarize the key contributions here because they supply both the raw material and the methodological templates for the spanning analysis in Section 6 and for the revealed-hedging-demand work we flag as future research.

Bürge et al. [2026] analyze over 300,000 Kalshi contracts and document that prices are broadly well-calibrated as probabilities and improve in accuracy as markets approach resolution, but display a clear favorite–longshot bias: low-price contracts win substantially less often than required to break even, while high-price contracts win slightly more often. They rationalize the pattern with a market-microstructure model distinguishing order-book makers (who earn a modest positive return, possibly with slight over-optimism) and takers (who lose roughly 32% on average).

Diercks et al. [2026] compare Kalshi’s macro contracts against Bloomberg surveys, Survey of Market Expectations data, and derivatives-implied forecasts, finding that Kalshi’s density forecasts update more frequently and sometimes more accurately than the alternatives. They note that Kalshi’s Fed funds rate contracts have attracted volumes comparable to SOFR-based instruments, outperform Fed funds futures forecasts for Fed decisions, and hypothesize that institutional participants with “greater hedging needs” may drive pricing patterns in these markets, suggesting but not formally testing a completion interpretation.

Eichengreen et al. [2025] provide the most direct methodological template for revealed-preference empirical work on prediction markets. They exploit Polymarket’s on-chain settlement—all trades are executed through publicly visible wallet addresses—to construct a panel of wallet-level positions across correlated contracts on Federal Reserve rate decisions and a contemporaneous contract on whether Chair Powell would be removed in 2025. They estimate a latent hawk–dove score for each wallet using an ideal-point model [Clinton et al., 2004]. Users whose trading revealed belief that Powell would be removed held more dovish short-rate expectations and expected higher long-term yields and inflation, consistent with reduced policy credibility.

Swanson et al. [2025] use Kalshi data around FOMC announcements to evaluate the “Fed information effect”; they find that contractionary monetary policy shocks reduce expectations for both growth and inflation, a pattern that is simpler to reconcile with a signaling interpretation of monetary surprises than with a conventional demand-channel interpretation.

Clinton and Huang [2025] provide an important counterweight to the prevailing optimism about prediction market accuracy. Analyzing more than 2,500 political contracts across Kalshi, PredictIt, Polymarket, and the Iowa Electronic Markets during the final five weeks of the 2024 presidential campaign, they report cross-platform accuracy rates of 93% (PredictIt), 78% (Kalshi), and 67% (Polymarket), and document substantial cross-venue price divergence for logically identical con-

tracts, negatively autocorrelated daily price changes, and arbitrage opportunities that peaked in the final two weeks before Election Day. They interpret these patterns as evidence that high-volume platforms attract herding and speculative noise rather than informational precision. These results provide a counterexample to the information aggregation argument in settings where one would expect the aggregative ability of prediction markets to be the most visible: high-attention, high-volume elections.

Whether this also weakens the completion argument is, we argue, a separate empirical question: a contract can provide valuable hedging services even if its price is a noisy probability forecast, provided that the contract’s payoff is correlated with agents’ consumption-relevant risks and that enough liquidity is available to execute hedges at modest transaction costs.

2.3 The Market-Completion Tradition

We now introduce an alternative approach to understanding prediction markets, the market completion tradition. The intellectual predecessor of the completion view is the Arrow–Debreu framework itself [Arrow, 1964, Debreu, 1959]. In its simplest two-period form, S states of the world obtain with positive probability; a full set of S Arrow securities—each paying one unit of numeraire in a distinct state—permits any agent to reshape consumption across states subject only to the intertemporal budget constraint. The first welfare theorem then delivers Pareto efficiency. When fewer than S linearly independent securities are available, markets are incomplete, and the resulting equilibrium allocations are generically suboptimal.

Magill and Quinzii [1996] provide a comprehensive treatment of economies with incomplete markets; Allen and Gale [1994] survey financial innovation through this lens. The most direct predecessor of this prediction-markets-as-completion argument is Robert Shiller’s research program on macro markets [Shiller, 1993, 2004, Athanasoulis and Shiller, 2001], which proposes exchange-traded contracts on GDP, real estate price indices, occupational income series, and other aggregate risks that individuals and institutions bear but cannot currently lay off. Athanasoulis and Shiller [2001] derives, for a set of candidate international macro securities, optimal contract design that maximizes welfare gains from risk-sharing.

Diercks et al. [2026] explicitly invoke the macro-markets framing in their title (“Kalshi and the Rise of Macro Markets”) and show that Kalshi’s contracts on CPI inflation, payrolls, unemployment, and Federal Reserve rate decisions yield well-calibrated, rapidly updating density forecasts of key macroeconomic variables. Their focus, consistent with the dominant literature, is on the forecasting quality of these prices and on their value as a benchmark for measuring macroeconomic expectations. The welfare question—what these contracts are worth to the agents who could use them to hedge—is left implicit. One contribution of the present paper is to make that question explicit.

2.4 Negative Results on Financial Innovation and Spanning Neutrality

Any serious completion argument must engage with a body of work showing that introducing additional securities into an incomplete market need not always improve welfare. [Hart \[1975\]](#) constructed early counterexamples in which the addition of a security reduces every agent’s utility. [Elul \[1995\]](#) and [Cass and Citanna \[1998\]](#) generalize these findings. [Geanakoplos and Polemarchakis \[1986\]](#) and a subsequent literature establish that equilibria of incomplete-market economies are generically constrained suboptimal: interventions that reshuffle the available asset span can improve or worsen welfare in ways that do not track obvious intuitions about spanning.

A separate and older strand, due to [Hirshleifer \[1971\]](#), makes a different but related point: public information released before the trading of state-contingent claims can inhibit insurance markets, because agents who would have been willing to insure one another ex ante lose the mutual gain once the state is revealed. Applied to prediction markets, this suggests that the aggregation function—which, by construction, reveals information—and the completion function can be in tension. A market that aggregates private information perfectly may thereby undermine the very risk-sharing the contract is designed to facilitate.

2.5 The Conflation in Applied Work

Despite the theoretical distinctions being well understood in pure theory, applied writing on prediction markets—including much of what appears in regulatory filings and policy briefs—treats aggregation and completion as aspects of a single phenomenon. [Arrow et al. \[2008\]](#), a widely cited *Science* essay signed by 22 prominent economists, argues for regulatory liberalization but articulates the case primarily in aggregation terms. Kalshi’s filings to the CFTC in the 2023–2025 litigation leaned more explicitly on hedging demand as a theory of economic purpose, but the underlying welfare economics are typically left informal. [Diercks et al. \[2026\]](#), even as they title their paper “Kalshi and the Rise of Macro Markets” and explicitly position the contracts within the Shiller tradition, focus almost entirely on the forecasting accuracy question. A principal goal of this paper is to sharpen the distinction so that the separate contributions of aggregation and completion can be measured, weighed, and regulated on their own terms.

3 Theory

3.1 Setup

Consider a two-period exchange economy with a finite state space $S = \{1, 2, \dots, s\}$ realized in period 1. There are I agents indexed by i . Each agent has a strictly concave, strictly increasing, twice differentiable von Neumann–Morgenstern utility function u^i over consumption in period 1, a common subjective probability distribution $\pi \in \Delta(S)$, and a state-contingent endowment $\omega^i \in$

$\mathbb{R}_{++}^s$.⁴ A complete set of Arrow securities consists of s assets, where asset j pays one unit of numeraire in state j and zero otherwise. Denote the price vector $q \in \mathbb{R}_{++}^s$.

With complete markets, agent i 's problem is

$$\max_{c^i \in \mathbb{R}_+^s} \sum_{j=1}^s \pi_j u^i(c_j^i) \quad \text{s.t.} \quad \sum_{j=1}^s q_j c_j^i \leq \sum_{j=1}^s q_j \omega_j^i.$$

The first welfare theorem applies: any competitive equilibrium allocation is Pareto optimal. In particular, equilibrium consumption c^{i*} satisfies the standard Borch condition that marginal utilities are proportional across agents and states, which in the case of common beliefs implies that individual consumption depends only on aggregate consumption in each state.

3.2 Prediction Markets as Arrow Securities

Now suppose the tradeable asset set is not the full collection of s Arrow securities but a strict subset spanned by $K < s$ linearly independent securities, with payoff matrix $A \in \mathbb{R}^{s \times K}$, where $A_{j,k}$ denotes the payoff of asset k in state j . Markets are incomplete. Let $E \subseteq S$ denote an event—a nonempty, strict subset of S —and define the indicator security $\mathbf{1}_E \in \mathbb{R}^s$ with $(\mathbf{1}_E)_j = 1$ if $j \in E$ and 0 otherwise. This is an exchange-traded binary prediction market contract on E .

Introducing $\mathbf{1}_E$ augments the asset span to the column span of $[A \mid \mathbf{1}_E]$. If $\mathbf{1}_E$ is not already in the span of A , the dimension of the attainable consumption set strictly increases for at least some agents.

Proposition 1 (Welfare gain from a spanning prediction contract). *Fix the incomplete-markets economy $(u^i, \omega^i, \pi)_{i=1}^I$ with asset matrix A . Suppose $\mathbf{1}_E \notin \text{col}(A)$ and there exist at least two agents i, i' such that $\lambda^i \cdot \mathbf{1}_E \neq \lambda^{i'} \cdot \mathbf{1}_E$, where $\lambda_s^i = \pi_s u^{i'}(c_s^{i*})/\mu^i$ is agent i 's state-price vector at the original equilibrium (equivalently, the two agents' marginal rates of substitution across E and $S \setminus E$ differ). Then there exists an equilibrium of the augmented economy $[A \mid \mathbf{1}_E]$ that Pareto dominates the original equilibrium.*

The proposition is a direct specialization of classical spanning results [see, e.g., [Magill and Quinzii, 1996](#), Ch. 10]; we include it to make the applied content explicit. The substantive content is that two primitives are required for welfare improvement: the new contract must add spanning (it must not be redundant), and agents must differ in the shadow value they place on the event (so that mutually beneficial trade exists). Both are stringent in principle but unrestrictive in practice: events of political, regulatory, or macroeconomic interest are almost never redundant relative to equities and bonds, and heterogeneous exposures across households and firms are pervasive.

The converse of Proposition 1 is also worth stating explicitly, because it delineates the empirical conditions under which prediction markets cannot improve welfare regardless of trading volume, price accuracy, or apparent demand.

⁴Heterogeneous beliefs complicate the completion analysis but do not affect its core logic; we maintain common beliefs for clarity of exposition and return to heterogeneity below.

Remark 1 (Spanning neutrality of binary contracts). *Fix the incomplete-markets economy $(u^i, \omega^i, \pi)_{i=1}^I$ with asset matrix A , and assume that $\mathbf{1} \in \text{col}(A)$ (a risk-free security is available).⁵ Suppose every agent’s endowment satisfies $\omega^i \in \text{col}(A)$ —equivalently, each ω^i can be replicated by some portfolio θ^i of the pre-existing assets. Then the competitive equilibrium of the economy with asset matrix A is Pareto efficient, and for any additional security x (in particular, any binary event contract $\mathbf{1}_E$), the competitive equilibrium consumption allocation and welfare level in the augmented economy $[A \mid x]$ coincide with those of A .*

The mechanism is immediate: under the stated condition, each agent can attain the Pareto-efficient consumption bundle by trading only within $\text{col}(A)$, so the shadow price of additional spanning at the equilibrium is zero. In particular, if A contains a bond and an equity whose payoff is a linear function of the state index z , and every agent’s endowment is also linear in z , then no binary prediction contract $\mathbf{1}_E$ has any welfare effect whatsoever. Non-linearity of agent exposure—concentrated losses or gains conditional on a particular subset of states—is a *necessary* condition for binary event contracts to deliver welfare gains. This observation is central to the calibration results of Sections 4–5.

3.3 Separating Market Completion from Information Aggregation

The modal claim about prediction markets concerns information: trading reveals probabilities that would otherwise remain private. We now illustrate that welfare gains from completion can arise even when no private information exists to be aggregated, which demonstrates the logical separation of the two rationales.

Suppose π is common knowledge; all agents agree on state probabilities. The only source of heterogeneity is the endowment vector ω^i . In this economy there is no private information, so the aggregation function is vacuous. Yet if an event E is correlated with $\omega^i - \omega^{i'}$ for at least some pair of agents—that is, if agents’ endowments differ across E and $S \setminus E$ —then the hypotheses of Proposition 1 are satisfied and introducing $\mathbf{1}_E$ strictly improves welfare. Intuition: a firm whose profits depend on a regulatory ruling, and a household whose wages do not, can share risk through $\mathbf{1}_E$ regardless of whether either party has any private information about the outcome.

The converse also holds. In a pure Grossman–Stiglitz economy with identical preferences, identical endowments, and heterogeneous private signals, prediction market prices aggregate information but the welfare gain in consumption terms is zero because agents have nothing to share. Aggregation and completion are therefore orthogonal in the sense that either can occur without the other.

3.4 Caveats and Negative Results

Two qualifications to the welfare claim must be taken seriously. The first, due to Hart [1975], is that Proposition 1 asserts the existence of a Pareto-dominating equilibrium in the augmented economy,

⁵This assumption is innocuous in applied settings: TIPS, money-market funds, or any nominal bond sufficiently short-dated that inflation erosion is negligible supplies a risk-free security.

not that all equilibria of the augmented economy dominate equilibria of the original. In incomplete-market economies with multiple equilibria, introducing a security can select an equilibrium that harms some or all agents. [Geanakoplos and Polemarchakis \[1986\]](#) show that this phenomenon is generic: constrained-suboptimal equilibria are the rule rather than the exception, and small perturbations to the asset span have ambiguous welfare effects.

The second, due to [Hirshleifer \[1971\]](#), is that the aggregation and completion functions can actively undermine one another. An Arrow security for event E permits ex ante risk-sharing across the partition $\{E, S \setminus E\}$. If, however, the price of the security reveals information that resolves the state before other risk-sharing markets clear, agents may lose access to mutual insurance they would otherwise have enjoyed. The practical force of this observation in contemporary prediction markets is limited—most resolve on political or regulatory timescales of weeks to months, during which substantial risk-sharing trade occurs—but the theoretical point remains that the two welfare channels are not additive in general.

A completion argument must therefore be conditional: prediction markets are welfare-improving when they span dimensions of real heterogeneity in agent exposures, when they do not select perverse equilibria in pre-existing incomplete markets, and when the information they reveal does not destroy pre-existing insurance. Each of these conditions is, as we argue below, empirically examinable.

3.5 Connection to Macro Securities

This framework places prediction markets within the broader category of “macro securities” proposed by [Shiller \[1993, 2004\]](#): exchange-traded contracts on aggregate economic risks that individuals and institutions bear but cannot easily hedge. Shiller’s GDP-linked securities, occupational income swaps, and real-estate index derivatives are exactly Arrow-like claims on coarse partitions of macroeconomic states. Binary prediction contracts generalize this idea in one direction (to events defined by arbitrary measurable functions of the state) and restrict it in another (to binary rather than continuous payoffs). The welfare analysis of [Athanasoulis and Shiller \[2001\]](#), which computes optimal security designs maximizing ex ante expected utility gains in a calibrated model, is directly portable: one can ask which events, among all those that could be listed, produce the largest welfare gain per unit of listing cost, and whether currently listed contracts resemble the welfare-optimal set. The calibration exercise in [Section 4](#) carries out this computation for the simplest case of a binary inflation contract and provides an order-of-magnitude estimate.

4 A Calibration Exercise for the Kalshi Inflation Contract

This section presents a minimal quantitative exercise designed to answer a concrete question: by how much, plausibly, does the introduction of a Kalshi-style binary contract on a macroeconomic event raise welfare through the completion channel alone? The exercise is deliberately simple—a two-agent, three-state, one-period economy—so that all steps can be solved transparently and the

results can be interpreted without recourse to the elaborate numerical machinery typical of full quantitative macro models. The exercise should be read as an existence proof (the welfare gains are not negligible) rather than a definitive calibration, and as a template that can be scaled up using standard tools.

4.1 Model Primitives

The state space is $S = \{L, M, H\}$, interpreted as low, medium, and high realizations of year-ahead core CPI inflation, with partition thresholds at 2% and 3% annual growth. State probabilities are calibrated to the empirical frequency of quarterly core CPI realizations in these bins over 1995Q1–2024Q4:

$$\pi = (\pi_L, \pi_M, \pi_H) = (0.35, 0.45, 0.20).$$

There are two representative agents. Agent H (“household”) represents nominal-income earners: retirees on fixed pensions, holders of long-duration nominal assets, workers with sticky nominal wages. Agent F (“firm”) represents inflation beneficiaries: firms with pricing power, owners of real assets, producers of price-flexible goods. State-contingent endowments are

$$\omega^H = (1.30, 1.00, 0.70), \quad \omega^F = (0.70, 1.00, 1.30),$$

so that aggregate endowment $\omega^H + \omega^F = (2, 2, 2)$ is flat across states—there is no aggregate risk, and first-best allocations deliver perfect risk-sharing. The only relevant heterogeneity is cross-sectional: agents bear opposite real-income exposures to the inflation state. This is a deliberate simplification. A richer calibration would permit aggregate fluctuations, heterogeneous risk aversion, and multiple agent types; the two-agent symmetric setting is chosen to produce clean closed-form complete-markets benchmarks.

Preferences are CRRA:

$$u^i(c) = \frac{c^{1-\gamma} - 1}{1-\gamma}, \quad \gamma > 0, \gamma \neq 1,$$

with $u^i(c) = \log c$ for $\gamma = 1$. The baseline calibration uses $\gamma = 2$.

The pre-existing asset menu consists of a nominal bond $B = (1, 1, 1)$ and an equity index S . To capture the empirical fact that equities are a poor inflation hedge—the Bodie–Fama–Schwert finding that real equity returns correlate negatively with inflation surprises [Bodie, 1976, Fama and Schwert, 1977], subsequently confirmed across asset classes [Fang et al., 2025]—we set $S = (1.00, 1.20, 0.80)$: equity performs best in moderate inflation and worst in high inflation. Both assets are assumed to be in zero net supply (agents take offsetting positions against one another); this is a standard simplification that isolates the consumption-reallocation role of financial markets. The span of $\{B, S\}$ contains only affine functions of the state index; in particular, the indicator payoff of any single state lies outside this span, so markets are incomplete.

The candidate new asset is the Kalshi-style inflation contract $I = \mathbf{1}_H = (0, 0, 1)$, which pays

one unit of numeraire if and only if inflation realizes in the high bin. Adding I to $\{B, S\}$ completes markets.

4.2 Equilibrium Concepts and Welfare Metric

We compute three equilibria:

1. Complete-markets benchmark, with all three Arrow securities available.
2. Incomplete-markets equilibrium, with only B and S tradeable.
3. Completed-markets equilibrium, with B , S , and I tradeable.

By symmetry of preferences and endowments and constancy of aggregate endowment, the complete-markets allocation has both agents consuming $\bar{c} = 1$ in every state, with equivalent Arrow prices $q_j^{\text{AD}} = \pi_j / \sum_k \pi_k = \pi_j$. The bond price is $q_B = 1$, the equity price $q_S = \pi_L + 1.2\pi_M + 0.8\pi_H = 1.05$, and the inflation-contract price $q_I = \pi_H = 0.20$.

Incomplete- and completed-market equilibria are solved numerically as constrained-planner problems with Negishi weight $\lambda = 1/2$, which by symmetry coincides with the zero-net-supply competitive equilibrium.⁶ Welfare is reported in certainty-equivalent consumption units $c^{i,\text{CE}}$ defined implicitly by $u^i(c^{i,\text{CE}}) = \mathbb{E}[u^i(c^i)]$, and the aggregate metric is $\sum_i c^{i,\text{CE}}$.

4.3 Results

Table 1 reports the baseline results at $\gamma = 2$. Under complete markets both agents achieve $c^{\text{CE}} = 1$. Under bond-and-equity incompleteness, household H 's certainty-equivalent consumption is 0.952 and firm F 's is 0.959, for an aggregate of 1.911—a 4.4% shortfall from the first-best benchmark of 2.000. Adding the inflation contract I restores both agents to the complete-markets allocation, delivering a Pareto improvement of +5.0% to household H and +4.2% to firm F , and an aggregate sum-certainty-equivalent gain of 4.6%. The pricing kernel gap between the two agents in the completed-markets equilibrium is on the order of 10^{-9} , confirming complete risk-sharing.

4.4 Sensitivity Analyses

Table 2 varies the coefficient of relative risk aversion $\gamma \in \{1, 2, 3, 5, 8\}$ with all other parameters held at baseline. Welfare gains from completion scale roughly linearly in γ : at $\gamma = 1$ (log utility) the aggregate gain is 2.3%; at $\gamma = 8$ it rises to 16.4%. This pattern is intuitive—more risk-averse agents value insurance more—and implies that empirical welfare estimates should be benchmarked against plausible ranges of γ drawn from the macro-finance literature (typically 2–5).

Table 3 varies the magnitude of heterogeneous inflation exposure, parametrized as $\omega^H = (1 + \eta, 1, 1 - \eta)$ and $\omega^F = (1 - \eta, 1, 1 + \eta)$. Welfare gains scale approximately quadratically in η ,

⁶We verify that the two agents' pricing kernels agree to numerical precision at the computed allocation, confirming that the planner's solution is a competitive equilibrium. Replication code is available on request.

Table 1: Baseline calibration: welfare under complete, incomplete, and completed markets ($\gamma = 2$, $\pi = (0.35, 0.45, 0.20)$).

	Complete benchmark	Incomplete (B, S)	Completed (B, S, I)
c^H in states (L, M, H)	(1.00, 1.00, 1.00)	(1.27, 0.88, 0.76)	(1.00, 1.00, 1.00)
c^F in states (L, M, H)	(1.00, 1.00, 1.00)	(0.73, 1.12, 1.24)	(1.00, 1.00, 1.00)
$c^{H,CE}$	1.000	0.952	1.000
$c^{F,CE}$	1.000	0.959	1.000
$c^{H,CE} + c^{F,CE}$	2.000	1.911	2.000
Gain over incomplete: H	—	—	+5.03%
Gain over incomplete: F	—	—	+4.24%
Aggregate CE gain	—	—	+4.63%
Prices (q_B, q_S, q_I)	(1.00, 1.05, 0.20)	(1.00, 1.04, —)	(1.00, 1.05, 0.20)

Table 2: Sensitivity to risk aversion (aggregate CE consumption, default π and endowment dispersion).

γ	$\sum_i c^{i,CE}$ incomplete	$\sum_i c^{i,CE}$ completed	Aggregate CE gain
1.0	1.956	2.000	+2.27%
2.0	1.911	2.000	+4.63%
3.0	1.869	2.000	+7.00%
5.0	1.796	2.000	+11.39%
8.0	1.718	2.000	+16.39%

consistent with standard local welfare approximations. At $\eta = 0.10$ (modest exposure, roughly one-tenth of average consumption shifting with the inflation state), the aggregate gain is only 0.49%; at $\eta = 0.50$ the gain reaches 14.5%. The empirical moment that disciplines the choice of η is the cross-sectional correlation between household real consumption and inflation realizations, which for U.S. households in CEX data tends to fall in the range 0.15–0.35 [Vanguard, 2023], consistent with gains of a few percentage points of certainty-equivalent consumption.

Table 3: Sensitivity to inflation-exposure heterogeneity η ($\gamma = 2$, default π).

η	$\sum_i c^{i,CE}$ incomplete	$\sum_i c^{i,CE}$ completed	Aggregate CE gain
0.10	1.990	2.000	+0.49%
0.20	1.961	2.000	+1.99%
0.30	1.911	2.000	+4.63%
0.40	1.841	2.000	+8.66%
0.50	1.747	2.000	+14.46%

Varying the high-inflation probability π_H from 0.05 to 0.40 (holding $\pi_M = 0.50$ and shifting mass between π_L and π_H) produces aggregate CE gains in the range 3.1%–4.6%, peaking near $\pi_H = 0.20$. The non-monotonicity reflects the fact that at extreme probabilities the high-inflation

state either contributes little to expected utility (when rare) or dominates beliefs to the point that consumption-smoothing has less value (when common); intermediate probabilities generate the largest unhedged dispersion in marginal utilities.

4.5 Interpretation and Caveats

Three lessons emerge. First, plausible welfare gains from a single well-chosen prediction contract are on the order of several percentage points of certainty-equivalent consumption, and thus are nontrivial. Because the toy economy is constructed with zero aggregate risk—inflation reshuffles consumption across agents but leaves the aggregate endowment flat—the 4.6% figure isolates the pure cross-sectional risk-sharing channel and should be read as an *upper bound* on the completion gain rather than a policy-relevant central estimate. Allowing the empirically dominant positive correlation between inflation and aggregate output would lower the marginal value of the inflation hedge and shrink the gain; the scaled calibration of Section 5, which embeds a realistic asset menu, delivers the substantially smaller figures we regard as policy-relevant. Second, the gains scale in directions that are empirically disciplinable: higher risk aversion, larger heterogeneous exposures, and intermediate event probabilities. Third, the gains hold even in an economy with zero aggregate risk, so they are not driven by any connection between prediction contracts and the aggregate stochastic discount factor—they are entirely about cross-sectional risk-sharing. This last point is important: it confirms the theoretical claim that the completion rationale is logically distinct from any aggregation-based pricing story.

Four caveats must be borne in mind. First, the exercise assumes agents can be identified as unambiguously “inflation-losers” or “inflation-winners,” whereas most real agents have more complex exposure profiles. Second, the asset span is deliberately restricted to bond and equity; in reality, inflation-linked bonds, commodity derivatives, and real estate already provide partial inflation hedges, which would reduce the marginal contribution of a binary contract. Third, transaction costs, position limits, and liquidity considerations are abstracted away; at current Kalshi liquidity levels these could materially reduce realized welfare gains, though Kalshi’s Fed funds contracts already reach volumes comparable to SOFR derivatives in some months [Diercks et al., 2026]. Fourth, the equilibrium analysis assumes the new asset is introduced into an otherwise stable equilibrium; the Hart–Geanakoplos–Polemarchakis caveat that completion can select adverse equilibria in incomplete-market economies with multiple equilibria is not operative here (the baseline economy has a unique equilibrium), but would need to be confronted in richer calibrations.

The exercise can be extended straightforwardly along several dimensions: multiple event partitions (contracts on CPI thresholds at 2%, 3%, 4% and so on), richer agent heterogeneity (multiple household types with different exposures, as in Vanguard’s “personalized inflation beta” framework). The next section carries out these extensions and reports a finding that qualitatively sharpens the theoretical argument: in a realistic asset menu that includes TIPS, the marginal welfare gain from adding Kalshi-style inflation threshold contracts depends almost entirely on the presence of agents whose exposures are *non-linear* in inflation. Smooth, linearly-exposed agents are

adequately hedged by TIPS alone.

5 A Scaled-Up Calibration with Heterogeneous Agents

The two-agent exercise of Section 4 was designed as an existence proof: welfare gains from completion can be material at standard calibrations. But its design was deliberately favorable to the binary contract—equity was specified as a poor inflation hedge and TIPS were absent from the asset menu. A more serious calibration should (i) use realistic cross-sectional agent heterogeneity drawn from empirical work on inflation exposure, (ii) include TIPS in the baseline asset menu, (iii) test the Kalshi-style contract’s *marginal* contribution conditional on TIPS and equity already being tradeable (and fully utilized), and (iv) examine how the result depends on the functional form of agents’ exposures. This section carries out that exercise. The headline finding is that the welfare case for inflation-threshold contracts survives but is substantially narrowed: linear-in-inflation exposures—the empirically dominant form of household inflation risk [Doepke and Schneider, 2006, Auclert, 2019]—are adequately hedged by TIPS, so prediction markets’ marginal contribution rests specifically on *non-linear* exposures such as debt covenants with inflation triggers, rental escalator clauses, and tax-bracket effects.

5.1 Agent Structure and Empirical Discipline

We partition the economy into six empirically-motivated agent types with smooth exposures to inflation, plus two types with threshold exposures. The smooth types follow the principal heterogeneity identified by Doepke and Schneider [2006]: net nominal positions, not income or expenditure shares, are the dominant source of cross-sectional welfare exposure to unexpected inflation.⁷ Retirees and other nominal-asset holders bear large positive exposure η (they are hurt by inflation); mortgagor households bear negative η (they benefit from the real erosion of their debt). Middle-class wage earners have moderate positive η (partial wage stickiness, per Auclert’s earnings-heterogeneity channel). Wealthy real-asset holders have small negative η . Firms with pricing power gain from inflation; firms without lose. Shares and exposures are reported in Table 4, where η_i denotes agent i ’s inflation-exposure coefficient in the endowment specification $\omega_i(s) = \text{share}_i \cdot \exp(-\eta_i z_s) / Z_i$ defined below (with z_s the standardized inflation state): a positive η_i means the agent’s real endowment falls as inflation rises, a negative η_i that it rises.

The two threshold-exposed types represent agents whose inflation risk is non-linear, as documented in the empirical corporate-finance and commercial-real-estate literatures. Covenant-trigger firms—roughly 23% of rated non-financial firms carry debt with covenants that tighten sharply in high-inflation macro states—take a concentrated loss if inflation crosses a threshold. Escalator-

⁷Doepke and Schneider [2006] find that the distributional consequences of a 10% unexpected inflation shock amount to transfers of 5–10% of net worth for heavily exposed cohorts, dwarfing the 0.28 percentage-point annual inflation-rate gap between top and bottom expenditure quintiles documented by the BLS in CPI-I data [Klick and Stockburger, 2021]. Cravino et al. [2020] confirm that CPI basket heterogeneity matters far less than wealth-portfolio heterogeneity for welfare consequences of inflation.

Table 4: Agent types in the scaled calibration: population share and inflation-exposure coefficient.

Type	Share	η	Interpretation
Retirees / nominal lenders	0.14	+0.30	Fixed-nominal-income exposure
Mortgagor households	0.19	−0.25	Real debt erosion
Middle-class wage earners	0.38	+0.10	Partial wage stickiness
Wealthy real-asset holders	0.14	−0.05	Real-asset exposure
Firms, high pricing power	0.05	−0.25	Pass-through gain
Firms, low pricing power	0.05	+0.15	Margin squeeze
Covenant-trigger firm	0.03	−25% revenue if $\pi > 3\%$	
Escalator-clause tenant	0.02	−30% income if $\pi > 4\%$	

clause tenants bear concentrated losses if inflation exceeds the trigger level in their commercial lease (typically 4%–5%). These agent types serve a diagnostic purpose: they represent a form of inflation risk that smooth hedges cannot fully span, and their behavior in the model isolates the distinctive contribution of binary event contracts.

For the smooth types, endowments are specified as $\omega_i(s) = \text{share}_i \cdot \exp(-\eta_i \cdot z_s) / Z_i$, where z_s is the standardized inflation state and $Z_i = \sum_s \pi_s \exp(-\eta_i z_s)$ normalizes so that $\mathbb{E}[\omega_i] = \text{share}_i$. We take z_s to be the π -standardized midpoints of the five inflation bins: with bin midpoints (0.5, 1.5, 2.5, 3.5, 4.5)% and probabilities π below, the implied mean is 2.42% and standard deviation 1.13 percentage points, giving the standardized state vector

$$z = (z_1, \dots, z_5) = (-1.70, -0.82, +0.07, +0.96, +1.84).$$

As a worked example, the retiree type ($\eta = +0.30$, share 0.14) has $Z_i = 1.045$ and endowment vector $\omega_i = (0.223, 0.171, 0.131, 0.101, 0.077)$ across the five states, which satisfies $\mathbb{E}[\omega_i] = 0.14$ by construction.⁸ For the threshold types, $\omega_i(s)$ takes a constant value for s below the trigger and a lower constant above. The state space is partitioned into five inflation bins ($< 1\%$, $1\text{--}2\%$, $2\text{--}3\%$, $3\text{--}4\%$, $> 4\%$) with probabilities (0.10, 0.28, 0.32, 0.20, 0.10) calibrated to quarterly core CPI over 1995–2024.

5.2 Asset Menu and Solution Method

The baseline asset menu contains a nominal bond B (paying approximately $1/(1+\pi_s)$ in real terms), an equity index E calibrated to the empirical inflation-beta of -0.35 per standard deviation [Fang et al., 2025], and TIPS paying a flat unit in every state. Kalshi-style binary contracts $\mathbf{1}\{\pi > k\}$ for $k \in \{1\%, 2\%, 3\%, 4\%\}$ are introduced incrementally. A final benchmark provides complete Arrow securities (one per state).

We solve the utilitarian Pareto-efficient allocation given each menu by maximizing $\sum_i \lambda_i \mathbb{E}[u_i(c_i)]$ subject to the budget constraint $c_i = \omega_i + A\theta_i$ and market clearing $\sum_i \theta_i = 0$, with equal Negishi weights $\lambda_i = 1/N$.⁹ The solver uses L-BFGS-B with analytic gradients on the reduced-form ($N -$

⁸The full set of η_i values in Table 4, mapped through this formula, reproduces the baseline endowment matrix used in Tables 5–9; replication code is available on request.

⁹The equal weights $\lambda_i = 1/N$ are a normalization of the planner problem and are *not* a claim that agents hold equal

1) K -dimensional parameter vector.

5.3 Results: The Welfare Frontier

Table 5 reports aggregate certainty-equivalent consumption under each asset menu, at $\gamma = 2$. Starting from autarky, adding the bond-equity-TIPS menu captures almost all available welfare: the sum of certainty-equivalent consumption rises from 0.9675 to 0.9992, a gain of 3.28% relative to autarky, or essentially all (about 99.7%) of the way from autarky to the complete-Arrow benchmark of 0.9993. Adding Kalshi threshold contracts on top of this menu yields a marginal gain of about 0.003% from the first contract ($\mathbf{1}\{\pi > 3\%\}$), with subsequent contracts adding negligibly; the cumulative gain from all four contracts is on the order of 0.01% ($0.9992 \rightarrow 0.9993$, where the displayed four-decimal change reflects rounding). Either way the Kalshi contribution is two orders of magnitude smaller than the bond-equity-TIPS contribution, and is concentrated almost entirely in the first threshold contract.

Table 5: Welfare frontier with 8-agent economy (6 smooth + 2 threshold-exposed), $\gamma = 2$. Percentage changes are computed from full-precision values; displayed entries are rounded to four decimals, so small marginal gains may not reconcile exactly from the rounded column.

Asset menu	$\sum_i c_i^{\text{CE}}$	Δ vs previous	Δ vs autarky
Autarky (no trading)	0.9675	—	—
Bond + Equity + TIPS	0.9992	+3.28%	+3.28%
+ Kalshi $\mathbf{1}\{\pi > 3\%\}$	0.9993	+0.003%	+3.28%
+ Kalshi $\{I_3, I_4\}$	0.9993	+0.0006%	+3.28%
+ all four Kalshi thresholds	0.9993	+0.0000%	+3.28%
Complete Arrow securities	0.9993	+0.0000%	+3.28%

The reason is the spanning argument of Remark 1. The smooth-type agents have endowments of the form $\omega_i(s) = a_i \cdot \exp(-\eta_i z_s)$, which for the empirical range of η is close to linear in z_s . The span of $\{B, E, T\}$ includes constants and one linear-in- z direction (equity), so it essentially contains the smooth-type exposures to first order—exactly the condition under which Remark 1 predicts zero Kalshi welfare contribution. The only exposures left unhedged are (i) second-order curvature in the smooth types’ endowments, and (ii) the first-order mass concentrated on a few states in the threshold types’ endowments. The former is quantitatively small (on the order of $\eta^2 \sigma_z^2 \approx 0.05$ for the largest η); the latter is what Kalshi contracts are designed to span, but the threshold types together account for only 5% of the population in this calibration, limiting their aggregate contribution. Section 5.7 below verifies this decomposition by replacing the exponential

equilibrium wealth; endowment shares are deliberately asymmetric (Table 4). The relevant claim is that the resulting allocation is the zero-net-supply competitive equilibrium. In a zero-net-supply economy assets only redistribute a fixed aggregate endowment, so the competitive equilibrium is the constrained Pareto optimum at which every agent’s state-price vector agrees on the traded span, $A^\top \lambda^i = q^*$ for all i . We verify this directly: computing each agent’s pricing kernel from the planner allocation via $q_k = \sum_s \pi_s u'_i(c_i(s)) A(s, k) / \sum_s \pi_s u'_i(c_i(s))$, the *maximum* deviation across all agents, assets, and menus reported in this section is below 10^{-7} , confirming that the planner solution satisfies each agent’s portfolio first-order conditions and is therefore the decentralized equilibrium.

parameterization with a linear one and confirming that Kalshi’s smooth-agent contribution collapses to exactly zero.

5.4 Isolating the Threshold-Exposure Channel

To isolate the contribution of threshold contracts, we rerun the welfare frontier on a restricted economy containing only the two threshold-exposed types, each normalized to unit expected endowment. Table 6 shows that in this restricted economy, adding Kalshi contracts $\mathbf{1}\{\pi > 3\%\}$ and $\mathbf{1}\{\pi > 4\%\}$ on top of the bond-equity-TIPS menu delivers a 0.31% gain in aggregate certainty-equivalent consumption—two orders of magnitude larger than the same contracts’ contribution in the full 8-agent economy, and exactly equal to the residual gap to the complete-markets benchmark.

Table 6: Welfare frontier in restricted economy with only the two threshold-exposed types (each rescaled so $\mathbb{E}[\omega_i] = 1$), $\gamma = 2$.

Asset menu	$\sum_i c_i^{\text{CE}}$	Δ vs autarky	Δ vs B+E+T
Autarky (no trading)	1.9714	—	—
Bond + Equity + TIPS	1.9729	+0.080%	—
+ Kalshi $\{I_3, I_4\}$	1.9791	+0.392%	+0.311%
Complete Arrow	1.9791	+0.392%	+0.311%

Two observations deserve emphasis. First, TIPS contribute almost nothing to threshold-exposed agents: the 0.08% bond-equity-TIPS gain over autarky is more than an order of magnitude smaller than the 3.28% gain in the smooth-agent economy. This is intuitive: the threshold agents’ exposure is concentrated on one or two states, and linear inflation hedges with support across all states cannot target the relevant risk. Second, once the Kalshi contracts $\mathbf{1}\{\pi > 3\%\}$ and $\mathbf{1}\{\pi > 4\%\}$ are added, the threshold agents can fully hedge their exposures and the complete-markets benchmark is attained exactly. The binary contracts are, quite literally, the Arrow securities these agents need to completely hedge their risk exposures.

5.5 How Much of the Economy Needs Threshold Hedging?

The obvious next question is empirical: what share of the aggregate economy has genuinely non-linear inflation exposures of the sort these two agents represent? Table 7 reports the welfare frontier as we scale up the threshold-exposed agent share from the baseline 5% to 80% of the population (holding their relative composition fixed, and compressing the smooth types proportionally). The marginal gain from adding all four Kalshi threshold contracts on top of bond-equity-TIPS rises from 0.004% at a 5% threshold-agent share to 0.94% at an 80% share. The relationship is increasing at an accelerating rate—approximately a power law with exponent near 1.5 (a log-log fit of gain on share yields a slope of about 1.5), not exponential.

The numbers can be put in context, but the relevant share is uncertain and the welfare estimate is correspondingly a range rather than a point. A rough aggregate of the sources cited above gives a sense of plausible magnitudes: roughly 23% of rated non-financial firms carry covenants that tighten

Table 7: Marginal Kalshi contribution on top of bond-equity-TIPS, as a function of the population share with threshold-type exposure ($\gamma = 2$).

Threshold-type share	$\sum c^{\text{CE}}$ autarky	$\sum c^{\text{CE}}$ B+E+T	$\sum c^{\text{CE}}$ + all I_k	B+E+T $\rightarrow$ Kalshi gain
0.025	0.9671	0.9993	0.9993	+0.0015%
0.050	0.9675	0.9992	0.9993	+0.0038%
0.100	0.9685	0.9991	0.9992	+0.0134%
0.200	0.9703	0.9984	0.9989	+0.0533%
0.400	0.9740	0.9956	0.9978	+0.2214%
0.800	0.9814	0.9836	0.9928	+0.9432%

in high-inflation states, CPI-escalator clauses are common in commercial leases (a sector that is on the order of a tenth of value added), and step-wise inflation adjustments appear in some pension and tax formulas. Netting these against overlap and against the fact that many such exposures are partially hedged by other means, an economy-wide threshold-exposed share somewhere in the 10–20% range is defensible, though we emphasize that no clean aggregate statistic exists and the mapping from these micro facts to a population share involves judgment. We therefore treat the 5% baseline of Table 4 as a deliberately conservative anchor and read welfare off the full sensitivity range in Table 7 (2.5% to 80%) rather than from any single share. At a 10–20% share, aggregate welfare gains from the full suite of Kalshi inflation-threshold contracts are on the order of 0.01–0.05% of certainty-equivalent consumption; applied to \$18 trillion of annual U.S. personal consumption expenditure, this is roughly \$2–10 billion per year. The figure is smaller than the toy-model magnitudes of Section 4 but is not zero, is disciplined—within the stated uncertainty—by the empirical literature on the heterogeneity of inflation exposure, and is large relative to the operating costs of the exchanges. Whether the realized gain reaches even this range depends on a separate, untested condition—that prediction markets actually reach the relevant exposed agents—which we take up in Section 5.9 and flag as an open empirical question rather than a maintained assumption.

5.6 Heterogeneity Intensity and the Approximation-Error Argument

A second way the scaled calibration generates non-trivial Kalshi contributions is through exposure-intensity scaling. Even with only smooth-type agents, increasing the exposure coefficient $\eta \rightarrow \kappa\eta$ for $\kappa > 1$ generates second-order curvature that linear assets cannot span. Table 8 reports the frontier under this scaling: at $\kappa = 1$ (baseline), Kalshi adds 0.0005% on top of bond-equity-TIPS; at $\kappa = 3$ (extreme heterogeneity) it adds 0.20%. The mechanism is that the endowment $\exp(-\kappa\eta z)$ develops meaningful Jensen’s-inequality curvature as κ grows, and this curvature is exactly what binary threshold contracts are positioned to hedge.

5.7 Robustness to the Functional Form of Exposure

The scaled calibration parameterizes smooth-agent endowments as $\omega_i(s) \propto \exp(-\eta_i z_s)$. This specific functional form is not essential to the conclusion. Remark 1 implies that what matters is

Table 8: Sensitivity to smooth-exposure intensity ($\eta \rightarrow \kappa\eta$), 6-agent smooth-only economy, $\gamma = 2$. This is the *without-threshold-agents* economy of Doepke and Schneider [2006]-style linear heterogeneity.

κ	$\sum c^{\text{CE}}$ autarky	$\sum c^{\text{CE}}$ B+E+T	$\sum c^{\text{CE}}$ + all I_k	B+E+T $\rightarrow$ Kalshi gain
0.5	0.9914	0.9999	0.9999	+0.00001%
1.0	0.9666	0.9993	0.9993	+0.00053%
1.5	0.9281	0.9976	0.9976	+0.00543%
2.0	0.8794	0.9938	0.9941	+0.02652%
3.0	0.7676	0.9762	0.9782	+0.20154%

whether each agent’s endowment lies within $\text{col}(A)$, which for our baseline asset menu $\{B, E, T\}$ contains the constants and linear-in- z functions exactly.¹⁰ Welfare gains from adding Kalshi contracts should therefore be driven by the component of agent endowments outside $\text{span}(\mathbf{1}, z)$ —that is, by second-order and higher-order curvature in z , or by discrete threshold effects—regardless of the specific functional form chosen.

We verify this prediction by rerunning the scaled calibration under three parameterizations of smooth-agent endowments: a linear form $\omega_i(s) \propto 1 - \eta_i z_s$ that by construction lies entirely within $\text{span}(\mathbf{1}, z)$; the exponential baseline; and a quadratic form $\omega_i(s) \propto 1 - \eta_i z_s + \kappa \eta_i^2 z_s^2$ that permits tunable curvature through the parameter κ ($\kappa = 0$ collapses to linear, $\kappa = 1/2$ matches the second-order Taylor expansion of the exponential, and $\kappa > 1/2$ amplifies the non-spanning curvature). All other primitives— η values, shares, state probabilities, threshold-agent specifications, and the asset menu—are held at the values of Section 5. Table 9 reports the results.

Table 9: Robustness of Kalshi’s marginal welfare contribution to the functional form of smooth-agent exposure, $\gamma = 2$. “Smooth only” is the 6-agent economy with threshold agents dropped (isolating the curvature channel). “Smooth + threshold” is the full 8-agent economy. The last column reports the L^2 norm of the projection residual of agent endowments onto $\text{span}(B, E, T)$, summed across the six smooth agents.

Smooth parameterization	Kalshi gain over B+E+T		Residual $\sum_i \ \omega_i - \text{Proj}\ _2$
	smooth only	smooth + threshold	
Linear $\omega_i \propto 1 - \eta_i z$	+0.00000%	+0.00321%	3.1×10^{-16}
Exponential $\omega_i \propto \exp(-\eta_i z)$ (baseline)	+0.00052%	+0.00380%	1.5×10^{-3}
Quadratic $\kappa = 0.5$ (matches exp to 2nd order)	+0.00001%	+0.00319%	2.0×10^{-4}
Quadratic $\kappa = 2.0$	+0.00004%	+0.00312%	7.3×10^{-4}
Quadratic $\kappa = 5.0$	+0.00014%	+0.00310%	1.6×10^{-3}

Three patterns deserve emphasis. First, under the linear parameterization the smooth-only Kalshi contribution is exactly zero to machine precision: the columns of $[B, E, T]$ span $\{1, z\}$, linear endowments lie entirely within this span, and Remark 1 applies verbatim. The numerical result confirms the analytic claim without ambiguity. Second, the marginal contribution of Kalshi

¹⁰TIPS delivers the constants. The equity payoff $E(s) = 1 - 0.35z_s$ delivers the linear-in- z direction. The bond payoff $B(s) = 1/(1 + \pi_s)$ is approximately constant with a small linear-in- z component, so $\{B, E, T\}$ spans $\{1, z\}$ to machine precision: the L^2 distance of z_s from its projection onto $\text{col}(B, E, T)$ is 1.7×10^{-15} .

contracts in the full 8-agent economy lies in a narrow band of $[0.00310\%, 0.00380\%]$ across all five parameterizations, because the threshold-agent channel dominates the smooth-curvature channel by one to two orders of magnitude in every case. The specific choice between linear, exponential, and quadratic smooth endowments moves the aggregate welfare number by less than 7×10^{-4} percentage points, which is a rounding error at the scale of the relevant policy question.¹¹ Third, the projection residual in the last column confirms the mechanism: linear endowments lie in $\text{col}(B, E, T)$ to machine precision, while exponential and larger- κ quadratic parameterizations place residual mass $\sim 10^{-3}$ outside the span, which is the mass that Kalshi contracts can in principle capture.

The robustness exercise also clarifies the role of agent count in the calibration. The six smooth agent types were introduced to make the calibration empirically interpretable: each type represents a recognizable segment of the population whose share and exposure can be disciplined by the empirical literature on heterogeneous inflation risk. But from a spanning perspective, what matters is the dimensionality of agent endowments relative to the asset payoff space, not the number of named types. One could collapse the six smooth types into two (a representative net lender and a representative net borrower) or expand them to thirty without changing the qualitative welfare conclusion, because a single dimension of linear-in- z heterogeneity behaves identically whether it is instantiated by two agents, six agents, or thirty. The threshold agents genuinely add new dimensions—their endowment vectors are not in $\text{span}(\mathbf{1}, z)$ —and are what drive Kalshi’s marginal contribution. The agent count is therefore a presentational choice; the welfare-relevant primitive is the span of endowments relative to the pre-existing asset menu.

5.8 Interpretation

Three lessons emerge from the scaled calibration, each sharpening or qualifying a claim that the toy model of Section 4 left at the level of existence-proof.

First, *the welfare case for Kalshi-style inflation-threshold contracts is not primarily a story about the total volume of inflation risk*. Total inflation risk in this economy is substantial (autarky-to-complete-markets welfare gain of 3.3%), but almost all of it is spanned by the existing menu of bond, equity, and TIPS. The marginal contribution of threshold contracts is about *what* residual risk looks like, not how much inflation risk exists in the first place.

Second, *the marginal contribution rests critically on the presence of non-linear exposures*. When all agents have linear exposures to inflation—whether smooth (bondholders, mortgagors) or strong—TIPS spans their hedging demand. Only when some agents have concentrated losses on a subset of states (covenant triggers, escalator clauses, pension kinks, tax-bracket effects) do binary event contracts add spanning value. This is consistent with the theoretical point of Proposition 1: a new security adds welfare only if it is not redundant relative to the existing span.

Third, *the implication for optimal contract design is that Kalshi-style exchanges should target*

¹¹Applied to \$18 trillion of U.S. annual personal consumption expenditure, this is a robustness band of roughly \$100 million per year—well within the uncertainty introduced by other calibration choices such as risk aversion and threshold-agent share.

categories of risk where non-linear exposure is pervasive. Inflation-threshold contracts are a candidate but probably a weak one given that linear inflation risk is already well-hedged by TIPS. Stronger candidates are events for which no existing derivative spans the relevant risk at all: regulatory rulings (FDA approvals, antitrust decisions), political outcomes (election results, labor union actions, Fed chair appointments), and catastrophic tail events in geographically specific weather or geopolitical risk.¹² The calibration exercise is thus a refinement rather than a rejection of the completion argument: it tells us where the completion argument is quantitatively large and where it is small.

A set of caveats, partially inherited from Section 4 and partially specific to the scaled calibration: the result that TIPS nearly completes markets for linear exposures depends on TIPS being a clean real bond with no basis risk, whereas actual TIPS have CPI-lag and basket mismatch issues that would leave more residual risk for binary contracts to hedge; the calibration assumes identical CRRA preferences across agent types, whereas empirical evidence supports heterogeneous risk aversion, which would enlarge the welfare consequences of incomplete spanning; and the exercise holds threshold triggers fixed at 3% and 4%, whereas an optimal contract-design exercise would choose trigger thresholds to maximize welfare subject to a given number of contracts.

5.9 Limited Asset Market Participation as an Amplifier

The calibration so far has assumed that every smooth-type agent trades the full $[B, E, T]$ asset menu. While this is the conventional assumption in incomplete-markets calibrations, it is empirically inaccurate. A large body of evidence documents that, in reality, most US households do not fully utilize the asset menu the calibration assumes. As such, we now extend the calibration to allow heterogeneous household asset access, and find that the welfare gain from inflation prediction contracts becomes substantially larger when the population that prediction markets can reach extends beyond the relatively narrow set of households that already trade TIPS.

We state the interpretation of this subsection up front, because the calibration that follows produces large dollar figures that are easy to misread. The two scenarios we compute are a *lower and an upper bound*, not central estimates. The lower bound (“Restricted”) assumes prediction markets inherit exactly the same participation barrier as TIPS; the upper bound (“Universal”) assumes they reach every non-TIPS household frictionlessly. The upper bound rests on an assumption—that Kalshi breaks through the participation barrier that excludes most households from inflation-protected assets—which the model cannot adjudicate and which we flag as the central open question for the empirical program. The paper’s baseline finding is unchanged by this extension: with a realistic asset menu that includes TIPS, the completion gain from inflation-threshold contracts is modest, on the order of 0.003–0.05% of certainty-equivalent consumption (roughly \$100M per year). The large numbers below describe how much *additional* welfare would be available *if* the access

¹²This is consistent with the observed composition of Kalshi’s product listings: as of early 2026, sports contracts account for approximately 70% of volume, with politics and macro roughly splitting the remainder [Bürgi et al., 2026]. Sports risk is unspanned by conventional financial assets and features many agents with non-linear exposure (team ownership, concessions, local hospitality), which fits the framework developed here.

assumption held, and should be read as a sensitivity to that assumption rather than as a forecast.

Empirical Motivation

Three pieces of evidence frame the participation problem.

Direct TIPS holdings are rare. The 2022 Survey of Consumer Finances reports that only 21% of US households hold any stocks directly, only 8.6% hold savings bonds. Direct holdings of corporate or mortgage-backed bonds fall to just 1.2% of households [SCF, 2022, St. Louis Fed, 2017]. Direct holdings of TIPS specifically are not surveyed in the SCF but, as a niche product within the already-thin Treasury-bond category, are almost certainly below 5% of households. The asset class that the spanning argument of Section 5 assumes to be universally available is, in fact, held directly by a small minority.

Indirect holdings via retirement accounts raise the effective rate but only partially. About 54% of US households have a retirement account; roughly 62% of US adults have some equity exposure through funds, 401(k)s, or IRAs [Gallup, 2025]. The TIPS market itself has roughly \$1.6 trillion outstanding, a substantial fraction of which sits in target-date and balanced funds. But indirect exposure mediated through fund-manager allocation decisions is not the same as a household actively choosing inflation protection; the household has, at most, exposure to the fund’s chosen blend, not the ability to titrate inflation hedging to their own non-linear endowment risk.

Non-participation is a 35-year-old puzzle. The empirical literature on limited stock-market participation dates to Mankiw and Zeldes [1991], with major contributions from Haliassos and Bertaut [1995], Vissing-Jørgensen [2002], Campbell [2006], and Gomes et al. [2021]. The explanations include fixed participation costs, financial illiteracy, ambiguity aversion, and behavioral frictions. These same frictions almost certainly apply to TIPS access. The fixed-cost explanation has particular force: a household for whom the time and cognitive cost of opening a TreasuryDirect account exceeds their expected inflation-hedging benefit will rationally not participate, and the bulk of US households evidently fall in this category.

Calibration Extension

We split each smooth-type agent i into a fraction p that participates fully in $[B, E, T]$ and a fraction $1 - p$ that holds only a savings-account-equivalent risk-free bond.¹³ Threshold-type agents (the paired tenant-landlord types from Section 5.4) retain full $[B, E, T]$ access regardless of p , since they represent stylized commercial entities for whom the household-finance participation friction does not bind.

We compare three scenarios at each participation rate:

- **Pre:** no Kalshi listed. Participants trade $[B, E, T]$; non-participants hold only the bond.

¹³This abstracts from heterogeneity in non-participants’ actual holdings. The 2022 SCF reports that 98% of US households have a transaction account; 52% have a retirement account. The “bond-only” restriction in the calibration approximates the latter group’s hedging options on inflation, since their equity exposure through retirement accounts mostly tracks the aggregate market with limited inflation-specific allocation choices.

- **Restricted:** Kalshi listed, but accessible only to current $[B, E, T]$ participants. This is the case where prediction markets inherit the same participation barrier as TIPS.
- **Universal:** Kalshi listed and accessible to all agents. Non-participants can now hedge inflation risk through Kalshi binaries, but they retain no access to bond, equity, or TIPS. This is the case where prediction markets break through the TIPS-level participation barrier.

The Universal scenario captures a specific feature of prediction markets that is often discussed informally but rarely formalized: the binary structure, mobile-app accessibility, low minimum trade size, and sportsbook-like interface of platforms like Kalshi may make them more easily accessible to households for whom TreasuryDirect remains inaccessible due to ease-of-access hurdles. The Restricted scenario captures the opposite possibility, where the same financial-literacy and fixed-cost barriers that exclude households from TIPS also exclude them from Kalshi.

We report two welfare metrics. The primary metric is the *endowment-share-weighted average certainty equivalent* (eswce), defined as

$$\text{eswce} = \sum_i \frac{\omega_i \cdot \pi}{\sum_j \omega_j \cdot \pi} \cdot \text{CE}_i,$$

where CE_i is agent i 's certainty-equivalent consumption and $\omega_i \cdot \pi \equiv \sum_s \omega_{i,s} \pi_s = \mathbb{E}[\omega_i]$ is agent i 's expected endowment.¹⁴ This metric is dollar-interpretable (it scales linearly with aggregate expected endowment) and avoids the redistribution-distortion that arises when summing unweighted certainty equivalents across agents with very different wealth levels. As a robustness check we also report $\sum_i E[u_i(c_i)]$, the planner's aggregate expected utility, which is monotone in the feasible set by construction.

Results

Table 10: Welfare gain from Kalshi inflation contracts under heterogeneous asset-market participation. Participants trade $[B, E, T]$; non-participants hold only a risk-free bond. Threshold agents always participate. Gains reported in endowment-share-weighted average certainty-equivalent consumption (eswce), expressed as percent change relative to the Pre scenario at the same participation rate.

Participation rate p	Restricted gain	Universal gain	Amplification
0.10	+0.00001%	+3.150%	342,639×
0.30	+0.00014%	+2.679%	18,866×
0.50	+0.00059%	+1.622%	2,768×
0.70	+0.00065%	+0.524%	812×
1.00	+0.00059%	+0.00059%	1×

¹⁴We weight by expected endowment share rather than calling this a “wealth-weighted” metric, because in our zero-net-supply economy assets only redistribute a fixed aggregate endowment and net financial wealth is zero in equilibrium; expected endowment share is then the natural notion of each agent's claim on aggregate resources. Because we normalize $\mathbb{E}[\omega_i] = \text{share}_i$ (Section 5), the metric reduces to the population-share-weighted average certainty equivalent $\sum_i \text{share}_i \cdot \text{CE}_i$.

Table 10 reports the headline numbers. Two patterns are worth describing in detail.

At $p = 1.0$ (full participation) the result reproduces Section 5.4. The Restricted and Universal scenarios coincide because there are no non-participants; the Kalshi gain is +0.00059%, consistent with the negligible full-participation contribution reported in Section 5.4. The aggregate welfare check (sum of expected utilities) gives the same monotonic ordering: +0.0011 utility units, identical in both scenarios.

At $p < 1$, the Restricted gain stays small while the Universal gain rises dramatically. The Restricted gain remains in the 0.00001% to 0.00065% range across all participation rates, because Restricted Kalshi only adds completion benefit for participants—a population that is already well-hedged by $[B, E, T]$ in the residual sense identified by Remark 1 and Section 5. The Universal gain rises sharply as p falls, because the non-participating population grows and Universal Kalshi extends inflation hedging to that population for the first time. At $p = 0.50$, Universal Kalshi delivers a +1.62% eswce gain, roughly 2,800 times larger than the Restricted gain at the same participation rate.

In dollar terms applied to US PCE of roughly \$18 trillion, the Universal gain at $p = 0.50$ corresponds to approximately \$292 billion per year, compared with about \$105 million per year for the Restricted gain. At $p = 0.30$ the Universal gain rises further, to roughly \$482 billion per year, while the Restricted gain remains at \$26 million per year. These are large numbers, but they are upper bounds contingent on the untested access assumption stated at the start of this subsection, not central estimates; they deserve interpretation rather than direct quotation.

The aggregate-welfare metric tells the same qualitative story without dollar normalization. At $p = 0.50$, the Restricted scenario raises aggregate expected utility by +0.0006 units while the Universal scenario raises it by +2.74 units—a more-than-four-thousand-fold difference. The two metrics agree on the ordering: Universal $\succ$ Restricted $\succ$ Pre at every participation rate, as required by feasibility.

The participation extension does not change the structural finding of Section 5: the welfare value of Kalshi-style inflation contracts, conditional on holding $[B, E, T]$, is modest. What changes is the relevant population. Under full participation, that population is essentially everyone with $[B, E, T]$ access, and the Kalshi gain is the order of \$100M per year when put into the context of yearly US PCE. Under realistic participation rates, the welfare-relevant population for Kalshi splits into two groups with very different unrealized welfare.

Participants, who already hold $[B, E, T]$, gain on the order of \$100M per year from adding Kalshi binaries to their menu. This is the welfare contribution that the Section 5.4 calibration measures.

Non-participants, who hold only a savings account, gain on the order of \$100B to \$500B per year *if* Kalshi reaches them. This is several orders of magnitude larger than the participant gain, because non-participants enter the calibration with essentially no inflation hedging at all—their effective $[B, E, T]$ spanning result is trivial.

Whether the non-participant gain is realized in equilibrium depends on whether prediction

markets actually break through the participation barrier that excludes households from TIPS. This is an empirical question that the model cannot answer. The Universal calibration is an upper bound, capturing the welfare available if Kalshi achieves frictionless reach. The Restricted calibration is a lower bound, capturing the welfare available if Kalshi inherits exactly the same friction as TIPS. The true value of the welfare gain lies in between these bounds and would be determined by the composition of Kalshi’s user base and the realized uptake of these types of inflation-hedging event contracts therein.

Two pieces of suggestive evidence point toward Kalshi having lower participation barriers than TIPS. First, the structure of binary inflation event contracts is objectively simpler than the duration-and-coupon structure of a Treasury security: “Will inflation exceed 3%?” is more easily understood than “Treasury Inflation-Protected Security maturing in seven years with 0.625% coupon adjusted by CPI-U with a three-month indexation lag.” Second, Kalshi’s user-acquisition channels run through smartphone applications and social media, which reach populations that retirement-account brokerages and traditional finance often do not. Whether these features translate into hedging-relevant participation is the question.

The Central Open Question

The participation extension places a single empirical question at the center of the welfare argument: among the agents who actually trade Kalshi inflation contracts, what fraction are trading because they have inflation exposure that nothing else hedges, and what fraction are trading for speculative or entertainment reasons unrelated to their real-economy exposure? If the former fraction is large, the Universal reading applies; if small, the Restricted reading does. Because the participation extension amplifies the welfare estimate by up to four orders of magnitude, the answer determines whether prediction markets are a marginal financial-product innovation or a major welfare-relevant development—and the two readings carry very different policy implications, the former calling for routine product approval, the latter for substantial regulatory attention to the inflation-hedging function of the platforms. A second, related measurement priority is to identify whether hedging traders are drawn from the TIPS-participating population or the non-participating population, since this is precisely what distinguishes the Restricted from the Universal column. Answering both questions requires wallet- or account-level position data.

Caveats

The model’s bond-only restriction for non-participants is sharper than reality. Many non-participants hold equity through retirement accounts, which provide partial linear-in- z inflation hedging. The Universal gain is therefore an upper bound on what Kalshi could deliver even under frictionless access. A more refined calibration with three groups (no-access, equity-through-retirement-only, and full- $[B, E, T]$) would produce intermediate numbers, but the qualitative ordering—that the welfare-relevant population for Kalshi is far larger than the existing TIPS-participating population—would survive.

A second caveat: the calibration treats the participation friction as exogenous. In reality, household participation decisions are partially endogenous to the cost and complexity of the asset menu. The same household that finds TreasuryDirect inaccessible may also find Kalshi inaccessible, simply because they do not trade financial instruments at all. The Universal calibration’s assumption that non-participants frictionlessly trade Kalshi is best interpreted as a policy-counterfactual rather than a positive prediction.

A third caveat: the calibration uses standard CRRA preferences, which understate the welfare value of insurance against extreme outcomes. Households with subsistence consumption thresholds, present-biased discounting, or other behavioral departures from expected-utility maximization may value Kalshi-style binary contracts more or less than CRRA implies. This is the standard limitation of all welfare calibrations in macro-finance, and the same caveat applies to TIPS, the bond, and complete-markets benchmarks.

6 Spanning Evidence

The theoretical separation of completion from aggregation, combined with the calibration suggesting non-trivial welfare gains, motivates an empirical program distinct from the existing literature on forecasting accuracy. The cleanest *direct* signature of completion would be revealed hedging demand—evidence that the agents who trade a prediction contract are those whose real-economy exposures correlate with its payoff—but that test requires wallet- or account-level position data and is beyond the scope of this paper; we flag it as the central direction for future work in Section 7. Here we implement the test the available data support: an asset-pricing *spanning* test, which identifies the necessary orthogonal-complement condition of Proposition 1 directly from market prices. As we emphasize below, spanning failure is a necessary but not sufficient condition for the completion channel to be active.

The classical asset-pricing approach to identifying “new” risk is the spanning test: regress returns on the candidate security against contemporaneous returns of existing tradeable instruments and ask whether the residual is non-zero. If the residual is statistically and economically meaningful, the candidate adds a hedging dimension that conventional markets do not span. This is the direct empirical analog of the spanning condition in Proposition 1: the contract is in the orthogonal complement of $\text{col}(A)$ if and only if its returns are not fully replicable by linear combinations of existing asset returns.

We implement this test on a panel of Kalshi binary contracts and three parallel specifications of the spanning basis: a yield-space basis that matches the asset-pricing convention of the empirical literature on inflation and policy expectations, an investable-ETF basis that approximates the asset menu an actual hedger would face, and a kitchen-sink basis that combines the two. Reporting all three on a common sample lets us identify which dimensions of conventional asset markets span which families of Kalshi contracts, and how robust the spanning conclusion is to specification choice.

Data. Our Kalshi panel covers 13,121 contract-day observations across 963 contracts and 163 events, spanning July 2021 through January 2026.¹⁵ The four contract families divide as follows: 338 Fed funds contracts (45 monthly events, \$19.2M cumulative notional), 214 CPI contracts (36 events, \$6.6M), 233 unemployment contracts (51 events, \$1.0M), and 178 payrolls contracts (31 events, \$2.4M).

The spanning regressors come from three sources. The two FRED daily series ΔDGS10_t (first difference of the 10-year nominal Treasury yield) and ΔT10YIE_t (first difference of the 10-year breakeven inflation rate, equal to $\Delta\text{DGS10}_t - \Delta\text{DFII10}_t$) supply the yield-space basis. The three ETF returns r_t^{TIP} , r_t^{IEF} , r_t^{SPY} — daily log returns on the iShares TIPS Bond ETF, the iShares 7–10 Year Treasury Bond ETF, and the SPDR S&P 500 ETF respectively — supply the investable-asset basis. Both bases are restricted to equity-market trading days, on which all five series are observed.

Specification. The dependent variable is the daily price change $\Delta p_{c,t} = p_{c,t} - p_{c,t-1}$ for contract c on date t . We use price changes rather than log returns because binary contracts are bounded on $[0, 1]$, which makes log returns unstable when prices approach the boundary; the robustness exercise reported below confirms that log returns deliver the same qualitative conclusions. The three specifications are estimated pooled within each contract family with cluster-robust standard errors at the event level (CR1 finite-sample correction):¹⁶

- (A) Yield-only: $\Delta p_{c,t} = \alpha + \beta_1 \Delta\text{DGS10}_t + \beta_2 \Delta\text{T10YIE}_t + \varepsilon_{c,t}$,
- (B) ETF-only: $\Delta p_{c,t} = \alpha + \gamma_1 r_t^{\text{TIP}} + \gamma_2 r_t^{\text{IEF}} + \gamma_3 r_t^{\text{SPY}} + \varepsilon_{c,t}$,
- (C) Kitchen sink: $\Delta p_{c,t} = \alpha + \gamma_1 r_t^{\text{TIP}} + \gamma_2 r_t^{\text{IEF}} + \gamma_3 r_t^{\text{SPY}} + \beta_1 \Delta\text{DGS10}_t + \beta_2 \Delta\text{T10YIE}_t + \varepsilon_{c,t}$.

We restrict the sample to contract-days with at least five trades to filter out illiquid quote-only observations, and to contracts with at least five surviving trading-day observations to ensure within-contract differencing is well-defined. The intersection of these filters with the trading-day calendar of the ETF panel produces a common sample of 4,354 contract-day observations across 306 contracts and 78 events, covering 2021-07-09 through 2026-01-23. By-family breakdown: 969 CPI observations across 16 events, 420 Unemployment across 15, 542 Payrolls across 15, and 2,423 Fed funds across 32.

The hypothesis from Proposition 1 is that the residual variance $\text{Var}(\varepsilon_{c,t})$ is large relative to $\text{Var}(\Delta p_{c,t})$. This would imply Kalshi contracts capture risk dimensions that the conventional asset space does not span, satisfying the necessary condition for the completion channel to be active. The competing hypothesis is high R^2 , which would indicate that Kalshi contracts merely repackage exposures already available through yields, ETFs, or both, in which case the welfare case would

¹⁵Daily closing prices, volumes, and bid–ask spreads for all CPI, Fed funds rate, monthly payrolls, and U.S. unemployment-rate binary contracts listed on Kalshi during this window. Source: Kalshi public trade-level releases. We aggregate to daily volume-weighted closes and apply the filters described in the text.

¹⁶Clustering at the event level is appropriate because contracts within the same event (the CPI menu for a single month, for example) respond jointly to news flow about the underlying event and have correlated residuals. Across events, contract returns are essentially independent.

rest entirely on the aggregation channel.

Headline result. Table 11 reports the R^2 of each specification by contract family. Three patterns emerge.

First, the spanning failure is overwhelming and robust across specifications. The pooled unexplained fraction is 95.7% under the yield-only specification, 95.1% under the ETF-only specification, and 94.9% under the kitchen sink. Among contract families, three of four (CPI, Unemployment, Payrolls) exceed 98% unexplained in every specification; only Fed funds drops materially below this, to 91.8% in the kitchen sink. The most comprehensive linear hedge portfolio one can build from publicly traded instruments leaves roughly 95% of Kalshi daily price variance unexplained.

Table 11: Spanning regressions of Kalshi daily price changes on three candidate sets of conventional asset-pricing factors. Estimated on a common sample of 4,354 contract-day observations spanning July 2021 to January 2026. Standard errors are cluster-robust at the event level. The unexplained fraction ($1 - R^2$) is the empirical analog of the orthogonal-complement dimension in Proposition 1.

Family	N	Events	R^2			Unexplained fraction		
			(A) yield	(B) ETF	(C) full	(A) yield	(B) ETF	(C) full
CPI	969	16	0.0043	0.0145	0.0146	99.6%	98.5%	98.5%
Unemployment	420	15	0.0026	0.0012	0.0052	99.7%	99.9%	99.5%
Payrolls	542	15	0.0175	0.0184	0.0200	98.2%	98.2%	98.0%
Fed funds	2,423	32	0.0682	0.0792	0.0816	93.2%	92.1%	91.8%
All contracts	4,354	78	0.0428	0.0489	0.0507	95.7%	95.1%	94.9%

Second, the ETF specification triples the CPI family’s R^2 relative to the yield-only specification (0.0145 versus 0.0043) and increases Fed funds R^2 by roughly 1.1 percentage points (0.0792 versus 0.0682). The CPI gain is the more substantively important: TIP and IEF returns jointly capture the breakeven-inflation channel through the duration-weighted spread between inflation-protected and nominal-bond total returns, while the pure breakeven yield $\Delta T10YIE$ omits the coupon and convexity effects that any actual investable inflation hedge carries. The Fed funds gain reflects equity-market risk (SPY) capturing risk-on/risk-off comovement around FOMC announcements, which yield changes alone cannot capture.

Third, adding yields on top of ETFs does almost nothing. The increment from the ETF-only to the kitchen-sink specification is +0.0018 pooled, with family-by-family gains in the +0.0001 to +0.0039 range. A formal F -test of the joint hypothesis $\beta_{\Delta DGS10} = \beta_{\Delta T10YIE} = 0$ in the kitchen-sink specification rejects only for Fed funds ($F_{2,2417} = 3.07$, $p = 0.046$) and pooled ($F_{2,4348} = 3.96$, $p = 0.019$); for CPI, Unemployment, and Payrolls the test fails to reject at conventional levels ($p = 0.96$, 0.44 , 0.65 respectively). Yields and ETFs are near-substitutes in spanning content for macro-event contracts, with ETFs slightly more comprehensive; for Fed funds, yields add a small statistically significant residual on top of ETFs.

Coefficient pattern. Table 12 reports the coefficient estimates. In the yield-only specification (panel A), CPI contracts load weakly positively on $\Delta T10YIE$ ($\hat{\beta}_2 = +0.18$, $t = +1.63$); the point estimate is non-zero in the expected direction, but statistically insignificant despite the breakeven inflation rate being the closest yield-space analog to a CPI hedge. Fed funds contracts load strongly positively on $\Delta DGS10$ ($\hat{\beta}_1 = +0.29$, $t = +8.63$), the expected sign: when the nominal yield rises, expected short-rates rise, and binary contracts on policy-rate thresholds appreciate. Payrolls contracts load on $\Delta DGS10$ in the same direction ($\hat{\beta}_1 = +0.09$, $t = +2.85$), consistent with a positive labor-market surprise lifting nominal yields and Kalshi prices jointly. Unemployment contracts load on nothing.

In the ETF-only specification (panel B), the coefficient pattern is informative and economically interpretable. CPI contracts load significantly positively on TIP ($+4.91$, $t = +3.15$) and significantly negatively on IEF (-3.85 , $t = -2.48$), the signature of a long-breakeven portfolio. The CPI Kalshi price moves with the duration-weighted TIP-minus-IEF spread, which is the investable analog of the breakeven yield. The point estimates are large in absolute terms because daily ETF returns are small relative to daily Kalshi price changes, but the ratio $-\hat{\gamma}_2/\hat{\gamma}_1 \approx 0.78$ is intuitive — it says a one-unit positive return on TIP and a 0.78-unit positive return on IEF roughly leave CPI Kalshi prices unchanged, which is what one would expect since these returns approximately offset in the breakeven dimension.

Fed funds contracts load significantly negatively on IEF (-3.55 , $t = -8.14$): when bond prices fall (yields rise), the implied policy-path price rises in sympathy, consistent with the policy expectation being the dominant driver of both. The Fed funds family also loads negatively on SPY (-0.69 , $t = -2.69$), reflecting risk-on/risk-off sensitivity around FOMC announcements. Payrolls contracts load on SPY ($+0.19$, $t = +2.88$), consistent with payrolls surprises that lift equity expectations also lifting unemployment-rate contract prices that pay on stronger labor-market outcomes. Unemployment contracts load on nothing: none of the three ETFs are individually significant.

In the kitchen-sink specification (panel C), severe multicollinearity (correlations of -0.97 between $\Delta DGS10$ and r^{IEF} and -0.79 between $\Delta DGS10$ and r^{TIP}) rotates coefficient mass between the duration-bearing ETFs and the corresponding yield change, producing individually insignificant ETF coefficients even where the joint span is preserved. The interpretable kitchen-sink result is that the marginal contribution of $\Delta T10YIE$ is significantly negative for Fed funds (-0.15 , $t = -1.97$) and pooled (-0.14 , $t = -2.33$): conditional on tradeable returns, a breakeven-inflation surprise slightly depresses Fed funds prices, a pattern consistent with the Fed reading higher expected inflation as a tightening signal that does not translate one-for-one into the Kalshi market’s short-rate expectation.

Why Unemployment fails to span at all. The complete absence of spanning for Unemployment contracts is striking. Among the four contract families, Unemployment is the one whose underlying release is least tightly coupled to the financial assets in our right-hand side. CPI releases move the breakeven inflation rate directly. FOMC decisions move the nominal yield curve

Table 12: Coefficient estimates from the three spanning specifications. Robust t -statistics in parentheses, clustered at the event level. Significance: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$. In panel C, only the yield additions are reported; the ETF coefficients shift relative to panel B because of multicollinearity with the yield regressors (see discussion below) and are difficult to interpret in isolation.

Panel A. Yield-only specification (A).

Family	Intercept	ΔDGS10	ΔT10YIE	R^2
CPI	−0.0059* (−1.87)	+0.001 (+0.02)	+0.180 (+1.63)	0.0043
Unemployment	+0.0019 (+0.68)	+0.021 (+0.25)	−0.119 (−0.93)	0.0026
Payrolls	+0.0026 (+1.25)	+0.093*** (+2.85)	+0.079 (+0.94)	0.0175
Fed funds	+0.0069*** (+3.28)	+0.294*** (+8.63)	−0.093* (−1.72)	0.0682
All contracts	+0.0034* (+1.93)	+0.234*** (+7.66)	−0.068 (−1.49)	0.0428

Panel B. ETF-only specification (B).

Family	Intercept	r^{TIP}	r^{IEF}	r^{SPY}	R^2
CPI	−0.0055* (−1.78)	+4.91*** (+3.15)	−3.85** (−2.48)	−0.42 (−1.43)	0.0145
Unemployment	+0.0016 (+0.53)	−1.05 (−0.48)	+0.94 (+0.69)	+0.04 (+0.13)	0.0012
Payrolls	+0.0028 (+1.39)	+0.25 (+0.22)	−1.59* (−1.94)	+0.19*** (+2.88)	0.0184
Fed funds	+0.0071*** (+3.41)	+0.40 (+0.61)	−3.55*** (−8.14)	−0.69*** (−2.69)	0.0792
All contracts	+0.0037** (+2.14)	+0.72 (+1.20)	−3.23*** (−8.12)	−0.44** (−2.28)	0.0489

Panel C. Kitchen-sink yield additions (C).

Family	ΔDGS10	ΔT10YIE	R^2
CPI	+0.02 (+0.14)	+0.02 (+0.15)	0.0146
Unemployment	+0.24 (+0.72)	−0.19 (−1.08)	0.0052
Payrolls	+0.10 (+0.39)	+0.07 (+0.66)	0.0200
Fed funds	+0.18 (+1.34)	−0.15** (−1.97)	0.0816
All contracts	+0.14 (+1.33)	−0.14** (−2.33)	0.0507

and risk-on/risk-off conditions directly. Payrolls releases move equity and yields jointly through their effect on growth expectations. Unemployment-rate level realizations, which are what Kalshi’s U3 contracts pay on, tend to be either largely anticipated through the contemporaneous payrolls release and continuing jobless claims, or orthogonal to the macro-asset news flow that drives ETF and yield-curve returns. The result that no asset — yield, ETF, or combined — spans Unemployment Kalshi prices is therefore the empirical realization, in this dataset, of Remark 1 in its starkest form: a contract on a state-contingent macroeconomic realization that no tradeable asset prices. If the completion-channel welfare argument operates anywhere in our data, it should operate most strongly here.

Yields and ETFs as near-substitutes. The near-zero increment from the ETF-only to the kitchen-sink specification has a clean interpretation. Yield changes are linearly equivalent (up to duration and convexity) to total returns on duration-matched bond portfolios, so the two right-hand sides span roughly the same column space in different basis representations. The economically meaningful distinction is that ETF returns also capture coupon income and non-linear convexity terms that pure yield changes omit, which accounts for the modest pooled R^2 improvement of the ETF specification over the yield specification. The kitchen-sink specification, which combines the two near-collinear bases, achieves R^2 only marginally above the ETF-only R^2 but at the cost of less interpretable individual coefficients. For applied purposes the ETF specification is preferable: it is the closer empirical proxy to the asset menu an actual hedger faces.

Connection to Proposition 1. The spanning test directly identifies the linear-algebraic condition $1_E \notin \text{col}(A)$ in Proposition 1. A finding of high unexplained fraction is the empirical realization of this condition. Across all four contract families and in every specification, the unexplained fraction is comfortably above 90%, and in three of four families above 98%. This is consistent with all four Kalshi contract families adding spanning to a conventional asset menu, whether that menu is defined by yield-curve constructs, by investable ETFs, or by both jointly.

The test does not, however, identify the heterogeneous-shadow-price condition (clause (ii) of Proposition 1), which requires wallet-level data on actual positions and trading. Spanning failure is thus a necessary but not sufficient empirical condition for the completion channel to be active.

The relationship between this section and the limited-participation calibration of Section 5.9 is worth sketching further. The participation calibration showed that the welfare value of Kalshi contracts depends on (i) the spanning property tested here and (ii) whether prediction markets reach the population of TIPS non-participants. The current spanning test addresses (i) and finds robust evidence consistent with the completion-channel hypothesis. A positive result on (ii) from a revealed-hedging-demand test would jointly support the Universal welfare reading in Table 10. A null result on (ii) would still leave the Restricted reading, which is bounded from below by the participant gain of roughly \$100M per year computed in Section 5.9.

Robustness: log returns and within-event R^2 . Two robustness specifications confirm the headline conclusion is not an artifact of specification choice. First, replacing the dependent variable with the Kalshi log return $\log(p_{c,t}/p_{c,t-1})$ gives pooled R^2 values of 0.0403 (yield-only), 0.0450 (ETF-only), and 0.0485 (kitchen sink), within a percentage point of the price-change results in Table 11. The qualitative spanning failure is unchanged across all three specifications.

Second, estimating each regression separately within each event—which absorbs event-level common shocks exactly as a full set of event fixed effects would, addressing the concern that pooled event-clustered standard errors do not control for shocks common to all contracts within a single CPI release or FOMC decision—yields median within-event R^2 values that are systematically higher than the pooled R^2 but still leave the spanning gap open. Table 13 reports medians across the 41 events with at least 20 contract-day observations in the common sample. The median pooled within-event R^2 under the kitchen sink is 16.3%, roughly triple the pooled across-event R^2 of 5.1%; controlling for contract-level and event-level heterogeneity thus raises explained variance materially, but still leaves 84% of within-event variance unexplained, confirming that the spanning failure is not an artifact of pooling across heterogeneous strikes or of cross-event drift. The within-event R^2 is higher than the pooled R^2 because cross-event idiosyncratic drift does not enter the within-event residual; the pattern says that ETF–Kalshi coupling is more visible when one holds the event fixed. But the median within-event unexplained fraction is still 84%, and is above 87% in every specification for every family. Spanning failure is robust to whether one defines residual risk at the contract-day or contract-within-event level.

Table 13: Median within-event R^2 across events with at least 20 contract-day observations in the common sample. Family-level cell sizes in parentheses indicate the number of events meeting the 20-observation threshold; Payrolls has no events meeting the threshold because the family’s event distribution is concentrated at small obs counts after the filter.

Family	(A) yield-only	(B) ETF-only	(C) kitchen sink
CPI ($n = 9$)	0.036	0.066	0.144
Unemployment ($n = 5$)	0.052	0.048	0.128
Payrolls ($n = 0$)	—	—	—
Fed funds ($n = 27$)	0.111	0.128	0.164
All contracts ($n = 41$)	0.086	0.107	0.163

The most direct tradeable hedge: Fed funds futures. The Fed funds family is the one family for which a near-perfect tradeable hedge for the underlying plausibly exists: CME 30-Day Fed Funds futures price the same realized policy rate that Kalshi’s rate-decision contracts pay on. If any conventional instrument were going to close a Kalshi spanning gap, it should be this one. We test the claim using the full front-through-sixth-month futures strip (FF1, . . . , FF6, Bloomberg FF1--FF6 Comdty daily settlement). Because Fed funds futures settle on the monthly-average effective rate, they are quoted as 100 minus the implied rate; a price increase corresponds to an expected-rate decrease, so a Kalshi contract paying on “rate exceeds threshold k ” should load

negatively on $\Delta(\text{FF price})$.

A naive front-month specification understates the futures’ relevance for a mechanical reason. The front-month contract settles on the current calendar month’s average rate, most of which is already realized by mid-month; its price therefore stops moving well before the contract expires, even as policy expectations shift. Empirically the front-month contract’s mean absolute daily price change over our window is roughly a fifth that of the sixth-deferred contract (0.7 versus 3.2 basis-point equivalents). The Kalshi rate contracts, by contrast, resolve at a future FOMC meeting that is on average 72 calendar days ahead in our sample (median 50, interquartile range 29–98). The economically appropriate hedge is therefore a *deferred* contract matched to the Kalshi contract’s resolution horizon, not the front month.

Table 14 reports the Fed funds family spanning R^2 under the existing specifications augmented with three Fed funds futures constructions: the front-month ΔFF1 , a fixed roughly-three-month-deferred ΔFF3 , and a horizon-matched regressor that selects, for each contract-day, the futures contract whose deferral best matches the Kalshi contract’s days-to-resolution.¹⁷

Table 14: Fed funds family spanning regressions augmented with Fed funds futures. Estimated on the 2,423 Fed-funds contract-day observations (32 events) of the common sample. Standard errors cluster-robust at the event level. The front-month FF1 has essentially zero explanatory power (an expiry-staleness artifact); the deferred FF3 and horizon-matched contracts add genuine but limited explanatory power.

Specification	N	R^2	Resid. SD	Unexpl. frac.
Yield-only (A)	2,423	0.0682	0.0799	93.2%
ETF-only (B)	2,423	0.0792	0.0794	92.1%
Kitchen sink (C)	2,423	0.0816	0.0793	91.8%
ΔFF1 (front) only	2,423	0.0000	0.0827	100.0%
ΔFF3 ($\sim$ 3-month) only	2,423	0.0654	0.0800	93.5%
ΔFF horizon-matched only	2,423	0.0476	0.0807	95.2%
Kitchen sink + ΔFF1	2,423	0.0826	0.0793	91.7%
Kitchen sink + ΔFF3	2,423	0.1278	0.0773	87.2%
Kitchen sink + ΔFF horizon-matched	2,423	0.1139	0.0779	88.6%

Three findings emerge, and together they sharpen the paper’s central argument.

First, the front-month futures has *zero* univariate explanatory power ($R^2 = 0.0000$) and its coefficient is wrong-signed and insignificant ($+0.003$, $t = +0.09$). This is the staleness artifact described above: the front contract is nearly inert by the time most Kalshi trading occurs. It illustrates why a careless implementation of the “most direct hedge” test would mislead.

Second, the deferred and horizon-matched contracts carry important, correctly-signed information. The ΔFF3 coefficient is -0.34 ($t = -4.59$) on its own and survives at -0.23 ($t = -3.10$, horizon-matched) in the kitchen sink, the expected negative sign. Adding ΔFF3 to the kitchen sink

¹⁷The matching rule routes days-to-resolution of ≤ 45 to FF1, ≤ 75 to FF2, ≤ 105 to FF3, ≤ 135 to FF4, ≤ 165 to FF5, and beyond to FF6. Because 45% of Fed funds contract-days fall within 45 days of resolution, the matched regressor inherits a large share of the inert front-month variation, which is why the fixed FF3 regressor outperforms it; see text.

raises Fed funds R^2 from 8.2% to 12.8% and is strongly significant ($F_{1,2416} = 128.1$, $p < 0.001$); the horizon-matched contract raises it to 11.4% ($F_{1,2416} = 88.1$, $p < 0.001$). The fixed FF3 regressor outperforms the horizon-matched one because the matcher routes 45% of observations to the inert front month; a more granular intra-month roll-adjusted strip would presumably do somewhat better, but the qualitative result is already clear.

Third, and most importantly, even the best tradeable policy-rate hedge leaves the Fed funds spanning gap mostly open. The kitchen-sink-plus-FF3 specification — which includes the breakeven-portfolio ETFs, both yield-curve constructs, and the deferred futures that prices the same underlying as the Kalshi contract — leaves 87.2% of Fed funds daily price variance unexplained. The single most hedgeable family in our panel, augmented with the single most direct tradeable instrument for its underlying, still fails the spanning test by a wide margin.

The interpretation is worth stating carefully. One might have expected the Fed funds family to be the place where the completion argument is *weakest*, precisely because policy-rate risk is hedgeable through futures. The finding that Fed funds futures explain only an additional 4–5 percentage points of variance — and that 87% remains unexplained even with futures in the regression — indicates that the Kalshi rate-decision contracts carry threshold-crossing and meeting-date-timing risk that the continuous futures price does not span. This is the asset-pricing analog of Remark 1’s distinction between linear-in-state exposure (which the futures hedges) and the non-linear, threshold-crossing exposure that a binary contract isolates. A futures position hedges the expected level of the policy rate; it does not hedge the discrete event “the FOMC raises by 25 bp at the March meeting,” which is what the binary contract pays on. The futures result therefore reinforces the completion interpretation: even where a near-ideal linear hedge exists, the binary contract spans risk the linear hedge cannot.

Refinement: constant-maturity futures. The horizon-matched futures regressor of Table 14 inherited a large share of inert front-month variation because 45% of Fed funds contract-days fall within 45 days of resolution. A cleaner construction is a constant-maturity series that interpolates the FF1–FF6 strip to a fixed deferral horizon each day, removing the within-month staleness cycle entirely.¹⁸ Table 15 reports the result. The constant-maturity series carry monotonically more spanning information as the horizon lengthens: the 90-day series alone delivers $R^2 = 0.166$ (versus 0.065 for the fixed FF3), and added to the kitchen sink it raises Fed funds R^2 to 0.184. All coefficients are correctly signed (negative) and strongly significant ($t = -8$ to -10). The monotone horizon pattern (R^2 of 0.113, 0.147, 0.166 for CM30, CM60, CM90 alone) is itself evidence that the Kalshi rate contracts track the deferred policy path rather than the near-realized front rate.

The substantive conclusion of the Fed funds futures exercise is unchanged and in fact sharpened: the best-engineered tradeable policy-rate hedge — a constant-maturity 90-day futures position — added to the full conventional asset menu still leaves 81.6% of Fed funds daily price variance

¹⁸For each date we approximate the deferral of contract FF*k* as the days from t to the midpoint of its settlement month, then linearly interpolate the strip price to a fixed 30-, 60-, or 90-day horizon. This is the standard constant-maturity roll used for volatility indices, adapted to the monthly Fed funds strip.

Table 15: Fed funds spanning with constant-maturity (CM) Fed funds futures. The CMH regressor is the daily change in the strip price interpolated to a fixed H -day deferral. Estimated on the 2,423 Fed funds contract-day observations of the common sample, event-clustered standard errors.

Specification	R^2	Resid. SD	Unexpl. frac.
Kitchen sink (no futures)	0.0816	0.0793	91.8%
CM30 only	0.1130	0.0779	88.7%
CM60 only	0.1468	0.0764	85.3%
CM90 only	0.1658	0.0755	83.4%
Kitchen sink + CM30	0.1543	0.0761	84.6%
Kitchen sink + CM60	0.1686	0.0755	83.1%
Kitchen sink + CM90	0.1839	0.0748	81.6%

unexplained. The single most hedgeable family in the panel, augmented with a purpose-built constant-maturity hedge for its underlying, remains far from spanned.

Robustness: lead-lag timing. A natural concern is that the high unexplained fraction reflects slow information transmission across venues rather than genuine spanning failure: if Kalshi prices lead or lag the conventional markets by a day, a contemporaneous regression understates the true co-movement. We test this by augmenting the ETF specification with one lead ($t + 1$) and one lag ($t - 1$) of every regressor. Table 16 reports the result. For the three macro-event families, lead and lag terms add negligibly to R^2 and the joint F -test on the six lead/lag coefficients fails to reject ($p = 0.27$ for CPI, 0.59 for Unemployment, 0.996 for Payrolls). For Fed funds the joint test rejects ($p < 0.001$), but the economic magnitude is trivial: R^2 rises only from 0.080 to 0.091. Even adding leads and lags of the CM90 futures on top of the full kitchen sink moves Fed funds R^2 only from 0.184 to 0.196. Slow cross-venue information transmission therefore explains at most about one percentage point of variance; the unexplained component is genuine orthogonal-complement risk, not a one-day-offset measurement artifact.

Table 16: Lead-lag timing check, ETF specification. “Contemp.” is the contemporaneous ETF regression; subsequent columns add one lag, one lead, or both, of each ETF regressor. The final column reports the joint F -test that the six lead and lag coefficients are zero.

Family	Contemp.	+Lag	+Lead	Full	F (lead + lag = 0)
CPI	0.0146	0.0181	0.0179	0.0223	$F_{6,\cdot} = 1.27, p = 0.27$
Unemployment	0.0010	0.0039	0.0076	0.0122	$F_{6,\cdot} = 0.78, p = 0.59$
Payrolls	0.0184	0.0186	0.0194	0.0196	$F_{6,\cdot} = 0.11, p = 0.996$
Fed funds	0.0803	0.0853	0.0860	0.0907	$F_{6,\cdot} = 4.60, p < 0.001$
All contracts	0.0494	0.0544	0.0523	0.0570	$F_{6,\cdot} = 5.88, p < 0.001$

Robustness: strike distance and the locus of unspanned risk. Remark 1 distinguishes linear-in-state exposure, which conventional assets span, from non-linear threshold-crossing exposure, which a binary contract isolates. For a binary contract, threshold-crossing risk is greatest

when the outcome is genuinely uncertain — when the price is near 0.50, the point of maximal Bernoulli variance — and smallest at tail strikes (price near 0 or 1). We stratify the ETF spanning regression by the contract’s distance from at-the-money, measured as $|p_{c,t-1} - 0.50|$ using the lagged price to avoid mechanical correlation with $\Delta p_{c,t}$, and split each family into at-the-money (ATM), mid, and tail terciles.

Table 17 reports the pattern. The spanning R^2 is *highest* at the money and *lowest* at the tails — the reverse of a naive reading of the remark. The resolution is in the denominator: ATM contracts have roughly six times the raw daily price-change variance of tail contracts (Fed funds: 0.0114 versus 0.0020). When a Fed funds contract is a near coin-flip, its large daily movements are partly explained by movements in rate expectations, giving $R^2 \approx 0.13$. When the contract is deep in a tail, it barely moves, and the little movement it has is idiosyncratic noise unrelated to the conventional markets, giving $R^2 \approx 0.03$. A continuous specification interacting the ETF regressors with the ATM-distance confirms the gradient (joint interaction $F_{3,2415} = 17.8$, $p < 0.001$ for Fed funds).

Table 17: ETF spanning R^2 by at-the-money-distance tercile. ATM = price nearest 0.50 (maximal threshold-crossing risk); Tail = price nearest 0 or 1. The final two columns report, for Fed funds, the R^2 when the CM90 constant-maturity futures regressor is added.

Family	ETF spanning R^2			Fed funds, +CM90		
	ATM	Mid	Tail	ATM	Mid	Tail
CPI	0.0269	0.0607	0.0196	—	—	—
Unemployment	0.0183	0.0143	0.0103	—	—	—
Payrolls	0.0352	0.0852	0.0325	—	—	—
Fed funds	0.1288	0.0709	0.0344	0.2319	0.2192	0.1220
All contracts	0.0823	0.0407	0.0248	—	—	—

The completion-relevant reading of this pattern is twofold. First, even in the stratum where conventional assets explain the most — at-the-money Fed funds contracts with the CM90 futures included, $R^2 = 0.232$ — the majority of price variance remains unspanned. Second, and more important, the *absolute* unspanned variance is concentrated precisely at the money: the ATM tercile combines the highest raw variance with an unexplained fraction still near 77%, so the dollar quantity of unhedgeable risk is largest exactly where the binary contract’s threshold-crossing payoff is most informative. Tail contracts have low R^2 not because they are well-hedged by conventional assets, but because they carry little risk of any kind. The risk that binary prediction contracts uniquely span — the threshold-crossing risk of a genuinely uncertain macroeconomic outcome — lives at the money, and it is there that the conventional asset menu leaves the most unhedged in absolute terms. This is the empirical content of Remark 1 read correctly: the relevant quantity is not the R^2 ratio but the absolute residual variance, and that is maximized in the ATM region where the non-linear payoff structure of the binary contract does its work.

7 Conclusion

The prevailing literature on prediction markets emphasizes their function as information aggregators. This framing is accurate as far as it goes, it is incomplete in a quite literal sense: it omits the welfare rationale that follows most directly from classical general-equilibrium theory, namely that binary event contracts are Arrow securities whose introduction expands the span of attainable state-contingent consumption and thus provide new, Pareto-dominant equilibria. The market-completion rationale is logically distinct from the aggregation rationale—either can obtain without the other—and is the rationale that most naturally supports the regulatory argument that prediction markets serve genuine economic purpose. A minimal toy calibration suggests that welfare gains from completion can be on the order of several percentage points of certainty-equivalent consumption for a single well-chosen contract, and a scaled-up calibration with empirically disciplined heterogeneous inflation exposures shows that these gains are substantial for agents with non-linear exposures (covenant triggers, escalator clauses, pension kinks) but are substantially attenuated when TIPS are already available. The completion argument does not defend prediction markets in general terms; it defends contracts that span risks not already hedged by the existing menu.

Developing this argument requires engagement, on the theoretical level, with the negative results showing that additional securities do not automatically improve welfare in incomplete markets, and with [Hirshleifer](#)’s observation that information production and risk-sharing can be substitutes rather than complements. Empirically, our spanning tests establish the *necessary* condition—roughly 95% of daily Kalshi price variance is unspanned by a comprehensive menu of investable hedges, and the gap survives even for the most hedgeable family augmented with its most direct futures contract—but they do not establish the *sufficient* condition that the agents trading these contracts are in fact hedging real exposures. The central direction for future work is therefore a test of revealed hedging demand from wallet- or account-level position data, of the kind [Eichengreen et al. \[2025\]](#) demonstrate is tractable on Polymarket’s on-chain ledger and that Kalshi’s CFTC-regulated large-trader filings could support.

The larger methodological recommendation is that researchers and policymakers should refuse the conflation of aggregation and completion. The two are welfare-relevant for different reasons, generate different empirical predictions, and admit different regulatory responses. Measuring them separately is a precondition for understanding what prediction markets are for, and for assessing the social value of the rapid regulatory liberalization currently underway. The scaled calibration further suggests that from the completion perspective the largest welfare opportunities lie not in inflation-threshold contracts—where TIPS already provides much of the available hedging—but in event categories where no existing derivative spans the relevant risk at all: regulatory rulings, political outcomes, and geographically-specific weather and geopolitical risk. Whether the composition of Kalshi’s actual contract listings resembles this welfare-optimal pattern is an open question that a rigorous optimal-contract-design exercise in the tradition of [Athanasoulis and Shiller \[2001\]](#), combined with the revealed-hedging-demand evidence discussed above, would be needed to

answer. As policy is fundamentally a question of welfare, it is imperative that we understand these markets in terms of their welfare improving capacity. Responsible governance and further empirical work on their benefits and shortcomings will enable market participants and policymakers to guide prediction markets towards forms that allow traders to efficiently hedge risk and attain more fully completed markets.

A Proof of Proposition 1

This appendix provides a proof of Proposition 1. The argument is a specialization of results in Magill and Quinzii [1996], Chapter 10, and is included here to make the mechanism transparent for an applied audience and to clarify exactly which genericity conditions are needed.

A.1 Setup and Existence

Fix the incomplete-markets economy $(u^i, \omega^i, \pi)_{i=1}^I$ with asset payoff matrix $A \in \mathbb{R}^{S \times K}$ of full column rank $K \leq S$. We maintain the standing assumptions of Section 3: strict concavity, strict monotonicity, and C^2 differentiability of each u^i , together with the Inada condition $\lim_{c \rightarrow 0^+} u^{i'}(c) = +\infty$ ensuring interior consumption; strict positivity of the endowment $\omega^i \in \mathbb{R}_{++}^S$; and strictly positive state probabilities $\pi \in \Delta(S)$. An agent trades a portfolio $\theta^i \in \mathbb{R}^K$ at asset prices $q \in \mathbb{R}^K$, subject to the zero-wealth budget constraint $q \cdot \theta^i = 0$ and the non-negativity constraint $c^i = \omega^i + A\theta^i \geq 0$. A competitive equilibrium is a tuple (c^*, θ^*, q^*) such that each agent maximizes $U^i(c^i) = \sum_s \pi_s u^i(c_s^i)$ subject to their budget, and markets clear ($\sum_i \theta^{*,i} = 0$). Existence follows from Magill and Quinzii [1996, Theorem 10.2].

We additionally assume throughout that $\mathbf{1} \in \text{col}(A)$, i.e., a constant payoff (e.g., TIPS, a short-dated nominal bond, or money) is tradeable. This assumption is innocuous in the applied settings we care about and simplifies the shadow-price argument below.

A.2 First-Order Conditions and State Prices

Interior consumption (from Inada) means each agent's equilibrium problem is characterized by first-order conditions alone. Writing $\mathcal{L}^i(\theta^i; \mu^i) = \sum_s \pi_s u^i(\omega_s^i + (A\theta^i)_s) - \mu^i(q \cdot \theta^i)$, differentiation with respect to θ_k^i yields

$$\sum_s \pi_s u^{i'}(c_s^{*,i}) A_{s,k} = \mu^i q_k \quad \text{for each } k = 1, \dots, K,$$

where $\mu^i > 0$ is the Lagrange multiplier on the budget constraint. Defining the (agent-specific) state-price vector $\lambda^i \in \mathbb{R}_{++}^S$ by $\lambda_s^i \equiv \pi_s u^{i'}(c_s^{*,i}) / \mu^i$, we obtain the compact form

$$A^T \lambda^i = q^* \quad \text{for each } i = 1, \dots, I. \tag{1}$$

Equation (1) says that all agents agree on how the *tradeable* assets are priced by their personal state-price vectors, but λ^i itself is not uniquely determined by q^* alone: only the image under A^T is pinned down. Decomposing $\mathbb{R}^S = \text{col}(A) \oplus \text{col}(A)^\perp$ with respect to the standard inner product and letting P_A denote orthogonal projection onto $\text{col}(A)$, we have $P_A \lambda^i$ uniquely determined by q^* (and therefore the same across agents), while $(I - P_A) \lambda^i$ can differ across agents in incomplete markets.

A.3 Shadow Prices of the Candidate Security

For any candidate new security $x \in \mathbb{R}^S$, define agent i 's shadow price as

$$p^i(x) \equiv \lambda^i \cdot x = \frac{\sum_s \pi_s u^{i'}(c_s^{*,i}) x_s}{\mu^i}.$$

This is the marginal willingness to pay of agent i for a unit of x at the original equilibrium. Decomposing $x = P_A x + (I - P_A)x$, we obtain

$$p^i(x) = \underbrace{\lambda^i \cdot P_A x}_{\text{same across } i} + \underbrace{\lambda^i \cdot (I - P_A)x}_{\text{varies across } i}.$$

The first term is common across agents because $P_A x \in \text{col}(A)$ and $A^T \lambda^i = q^*$ for all i ; explicitly, if $P_A x = A\xi$ for some $\xi \in \mathbb{R}^K$, then $\lambda^i \cdot A\xi = (A^T \lambda^i) \cdot \xi = q^* \cdot \xi$. The second term reflects agents' heterogeneous valuations of the component of x outside the existing asset span.

If $x \in \text{col}(A)$, then $(I - P_A)x = 0$ and all agents agree on $p^i(x)$ —this is the standard Law of One Price for assets already spanned. If $x \notin \text{col}(A)$, there is scope for shadow prices to differ. Whether they actually differ depends on whether agents' excess marginal utilities $(I - P_A) \lambda^i$ have distinct projections onto x .

The hypothesis of Proposition 1 is precisely that two agents i, i' exist with $p^i(\mathbf{1}_E) \neq p^{i'}(\mathbf{1}_E)$; equivalently, with different marginal rates of substitution across E and $S \setminus E$ at the original equilibrium consumption.

A.4 Pareto Improvement via Heterogeneous Shadow Prices

Without loss of generality, suppose $p^i(\mathbf{1}_E) > p^{i'}(\mathbf{1}_E)$. Choose any $\hat{p} \in (p^{i'}(\mathbf{1}_E), p^i(\mathbf{1}_E))$. We construct a feasible allocation that Pareto-dominates the original equilibrium allocation.

Let $\varepsilon > 0$ be small. Consider the following perturbation to consumption:

$$\begin{aligned} \Delta c^i &= \varepsilon \mathbf{1}_E - \varepsilon \hat{p} r \\ \Delta c^{i'} &= -\varepsilon \mathbf{1}_E + \varepsilon \hat{p} r \\ \Delta c^j &= 0 \quad \text{for } j \notin \{i, i'\} \end{aligned}$$

where $r \in \text{col}(A)$ is any constant-payoff portfolio return (which exists by the standing assumption

$\mathbf{1} \in \text{col}(A)$). Agent i has effectively purchased ε units of $\mathbf{1}_E$ from agent i' at price $\hat{p}$ per unit, with the payment routed through a pre-existing risk-free asset. Aggregate resources are conserved: $\sum_j \Delta c_s^j = 0$ for each state s .

Feasibility requires $c^{*,j} + \Delta c^j \geq 0$ for each j , which holds for sufficiently small ε by the strict positivity of equilibrium consumption.

The first-order utility change for agent i is

$$\Delta U^i = \sum_s \pi_s u^{i'}(c_s^{*,i})(\Delta c_s^i) + O(\varepsilon^2) = \mu^i \varepsilon (p^i(\mathbf{1}_E) - \hat{p}) + O(\varepsilon^2) > 0$$

for $\varepsilon > 0$ small. By the same calculation, $\Delta U^{i'} = \mu^{i'} \varepsilon (\hat{p} - p^{i'}(\mathbf{1}_E)) + O(\varepsilon^2) > 0$. Other agents are unchanged. Hence the perturbed allocation strictly Pareto-dominates the original.

A.5 Equilibrium Implementation

The Pareto-improving perturbation in A.4 is an *allocation*, not an equilibrium. To conclude the proof, we need to show that an equilibrium of the augmented economy $[A \mid \mathbf{1}_E]$ Pareto-dominates the original equilibrium.

Existence of an equilibrium $(c^{**}, \theta^{**}, q^{**})$ in the augmented economy follows from [Magill and Quinzii \[1996, Theorem 10.2\]](#), applied to the asset matrix $[A \mid \mathbf{1}_E]$ of full column rank $K + 1$.

Pareto dominance at the new equilibrium follows from the generic Pareto-improvement result of [Magill and Quinzii \[1996, Theorem 10.8\]](#): for a generic (open, dense, full-measure) set of endowment profiles $\{\omega^i\}_{i=1}^I$ in $\mathbb{R}_{++}^{SI}$, any equilibrium of $[A \mid \mathbf{1}_E]$ strictly Pareto-dominates any equilibrium of A whenever $\mathbf{1}_E \notin \text{col}(A)$ and agents' shadow prices for $\mathbf{1}_E$ are heterogeneous at the original equilibrium.

The genericity qualifier in the Magill-Quinzii theorem is a response to the [Hart \[1975\]](#) counterexamples: for a Lebesgue-measure-zero set of endowment profiles, the introduction of a new security can leave some agents worse off due to price-level effects that offset the direct spanning gain. The hypothesis of Proposition 1—that shadow prices differ at the original equilibrium—excludes the most obvious degenerate cases but does not rule out all of them. For applied purposes the distinction is immaterial: the set of endowments where Pareto dominance fails is nowhere dense and has zero Lebesgue measure, so perturbations of any calibrated economy will satisfy the improvement claim with probability one.

This completes the proof. ■

A.6 Remarks

On the role of the common-constant-payoff assumption. The standing assumption $\mathbf{1} \in \text{col}(A)$ was used in A.4 to allow agents to pay for units of $\mathbf{1}_E$ via a constant-payoff instrument. Without this assumption the argument goes through with minor modifications: the price $\hat{p}$ is paid in some other tradeable asset, and the first-order argument proceeds identically. The cleaner statement with a

tradeable risk-free security is preferred because it matches all applied settings of interest.

On the role of heterogeneity in shadow prices. If all agents have identical shadow prices at the original equilibrium—which can occur by coincidence even in generic economies, but is a measure-zero event—then the introduction of $\mathbf{1}_E$ is a neutral perturbation: no agent wants to trade. The proposition’s hypothesis that shadow prices differ is therefore the substantive content; the rest of the proof is mechanical.

Relationship to Remark 1. Remark 1 is the contrapositive: if every agent’s endowment lies in $\text{col}(A)$, then their shadow prices λ^i agree on $\text{col}(A)^\perp$ (because the equilibrium consumption is already the unconstrained optimum, which coincides across agents up to endowment differences already spanned by A). In that case the heterogeneous-shadow-price hypothesis of Proposition 1 fails, and no Pareto improvement is available. The remark identifies the exact boundary at which the proposition’s welfare-gain conclusion turns off.

References

- Allen, F., and D. Gale (1994). *Financial Innovation and Risk Sharing*. MIT Press.
- Arrow, K. J. (1964). “The Role of Securities in the Optimal Allocation of Risk-Bearing.” *Review of Economic Studies* 31(2): 91–96.
- Arrow, K. J., R. Forsythe, M. Gorham, R. Hahn, R. Hanson, J. O. Ledyard, S. Levmore, R. Litan, P. Milgrom, F. D. Nelson, G. R. Neumann, M. Ottaviani, T. C. Schelling, R. J. Shiller, V. L. Smith, E. Snowberg, C. R. Sunstein, P. C. Tetlock, P. E. Tetlock, H. R. Varian, J. Wolfers, and E. Zitzewitz (2008). “The Promise of Prediction Markets.” *Science* 320(5878): 877–878.
- Athanasoulis, S. G., and R. J. Shiller (2001). “World Income Components: Measuring and Exploiting Risk-Sharing Opportunities.” *American Economic Review* 91(4): 1031–1054.
- Auclert, A. (2019). “Monetary Policy and the Redistribution Channel.” *American Economic Review* 109(6): 2333–2367.
- Berg, J. E., F. D. Nelson, and T. A. Rietz (2008). “Prediction Market Accuracy in the Long Run.” *International Journal of Forecasting* 24(2): 285–300.
- U.S. Bureau of Economic Analysis (2024). “Value Added by Industry: Real Estate and Rental and Leasing as a Percentage of GDP.” FRED series VAPGDPRL, retrieved from Federal Reserve Bank of St. Louis.
- Bodie, Z. (1976). “Common Stocks as a Hedge Against Inflation.” *Journal of Finance* 31(2): 459–470.
- Bürgi, C., W. Deng, and K. Whelan (2026). “Makers and Takers: The Economics of the Kalshi Prediction Market.” CEPR Discussion Paper No. 20631.

- Campbell, J. Y. (2006). “Household Finance.” *Journal of Finance* 61(4): 1553–1604.
- CBRE Research (2024). “U.S. Cap Rate Survey H2 2024.” CBRE Econometric Advisors, December.
- CARR Healthcare Real Estate (2024). “The Downfalls of CPI Escalations in Commercial Leases.” Industry advisory, August. [Earlier version: CESifo Working Paper No. 12122, September 2025.]
- Cass, D., and A. Citanna (1998). “Pareto Improving Financial Innovation in Incomplete Markets.” *Economic Theory* 11(3): 467–494.
- Clinton, J., S. Jackman, and D. Rivers (2004). “The Statistical Analysis of Roll Call Data.” *American Political Science Review* 98(2): 355–370.
- Clinton, J. D., and T. Huang (2025). “Prediction Markets? The Accuracy and Efficiency of \$2.4 Billion in the 2024 Presidential Election.” Working paper, Vanderbilt University.
- Commercial Property Connect (2022). “My Landlord Wants my Lease Tied to CPI?” Industry advisory, March.
- Cravino, J., T. Lan, and A. A. Levchenko (2020). “Price Stickiness along the Income Distribution and the Effects of Monetary Policy.” *Journal of Monetary Economics* 110: 19–32.
- Debreu, G. (1959). *Theory of Value*. Wiley.
- Diercks, A. M., J. D. Katz, and J. H. Wright (2026). “Kalshi and the Rise of Macro Markets.” Finance and Economics Discussion Series 2026-010, Federal Reserve Board; NBER Working Paper No. 34702.
- Doepke, M., and M. Schneider (2006). “Inflation and the Redistribution of Nominal Wealth.” *Journal of Political Economy* 114(6): 1069–1097.
- Eichengreen, B., G. Viswanath-Natraj, J. Wang, and Z. Wang (2025). “Under Pressure? Central Bank Independence Meets Blockchain Prediction Markets.” SSRN Working Paper 5366862.
- Elul, R. (1995). “Welfare Effects of Financial Innovation in Incomplete Markets Economies with Several Consumption Goods.” *Journal of Economic Theory* 65(1): 43–78.
- Fama, E. F., and G. W. Schwert (1977). “Asset Returns and Inflation.” *Journal of Financial Economics* 5(2): 115–146.
- Fang, X., Y. Liu, and N. Roussanov (2025). “Getting to the Core: Inflation Risks Within and Across Asset Classes.” *Review of Financial Studies*, forthcoming; NBER Working Paper No. 30169.
- Gallup News. “What Percentage of Americans Own Stock?” May 5.

- Geanakoplos, J. D., and H. M. Polemarchakis (1986). “Existence, Regularity, and Constrained Suboptimality of Competitive Allocations When the Asset Market Is Incomplete.” In W. P. Heller, R. M. Starr, and D. A. Starrett (eds.), *Uncertainty, Information and Communication: Essays in Honor of Kenneth J. Arrow, Volume III*. Cambridge University Press, 65–95.
- Gomes, F., M. Haliassos, and T. Ramadorai (2021). “Household Finance.” *Journal of Economic Literature* 59(3): 919–1000.
- Gomez Cram, R., Y. Guo, and T. Jensen, and H. Kung (2026). ”Prediction Market Accuracy: Crowd Wisdom or Informed Minority?” SSRN Working Paper 6617059
- Grossman, S. J. (1976). “On the Efficiency of Competitive Stock Markets Where Traders Have Diverse Information.” *Journal of Finance* 31(2): 573–585.
- Grossman, S. J., and J. E. Stiglitz (1980). “On the Impossibility of Informationally Efficient Markets.” *American Economic Review* 70(3): 393–408.
- Haliassos, M., and C. C. Bertaut (1995). “Why Do So Few Hold Stocks?” *Economic Journal* 105(432): 1110–1129.
- Hanson, R. (2003). “Combinatorial Information Market Design.” *Information Systems Frontiers* 5(1): 107–119.
- Hanson, R. (2007). “Logarithmic Market Scoring Rules for Modular Combinatorial Information Aggregation.” *Journal of Prediction Markets* 1(1): 3–15.
- Hart, O. D. (1975). “On the Optimality of Equilibrium When the Market Structure Is Incomplete.” *Journal of Economic Theory* 11(3): 418–443.
- Hayek, F. A. (1945). “The Use of Knowledge in Society.” *American Economic Review* 35(4): 519–530.
- Hirshleifer, J. (1971). “The Private and Social Value of Information and the Reward to Inventive Activity.” *American Economic Review* 61(4): 561–574.
- KalshiEx LLC v. Commodity Futures Trading Commission*, U.S. District Court for the District of Columbia, Civil Action No. 23-3257 (JMC), Memorandum Opinion and Order, September 12, 2024.
- Klick, J., and A. Stockburger (2021). “Experimental CPI for Lower and Higher Income Households.” U.S. Bureau of Labor Statistics Working Paper 537.
- Magill, M., and M. Quinzii (1996). *Theory of Incomplete Markets*. MIT Press.
- Mankiw, N. G., and S. P. Zeldes (1991). “The Consumption of Stockholders and Nonstockholders.” *Journal of Financial Economics* 29(1): 97–112.

- Manski, C. F. (2006). “Interpreting the Predictions of Prediction Markets.” *Economics Letters* 91(3): 425–429.
- Newmark Research (2022). “Escalation Nation: Rent isn’t the Only Industrial Occupancy Cost Increasing.” Industrial Insight, October.
- Norris McLaughlin, P.A. (2022). “Client Alert—Commercial Leases: CPI Revisited.” Legal advisory, March.
- Lewandowski, B., A. Illig, E. Street, and R. Wobbekind (2025). “Economic Impacts of Commercial Real Estate, 2025 U.S. Edition.” NAIOP Research Foundation, Leeds School of Business, University of Colorado Boulder, January.
- Ottaviani, M., and P. N. Sørensen (2015). “Price Reaction to Information with Heterogeneous Beliefs and Wealth Effects: Underreaction, Momentum, and Reversal.” *American Economic Review* 105(1): 1–34.
- REIPrime Glossary (2025). “CPI Adjustment in Real Estate Leases.” Industry reference, November.
- Board of Governors of the Federal Reserve System (2023). “Changes in U.S. Family Finances from 2019 to 2022: Evidence from the Survey of Consumer Finances.” *Federal Reserve Bulletin*, October.
- Shiller, R. J. (1993). *Macro Markets: Creating Institutions for Managing Society’s Largest Economic Risks*. Oxford University Press.
- Shiller, R. J. (2004). *The New Financial Order: Risk in the 21st Century*. Princeton University Press.
- Snowberg, E., J. Wolfers, and E. Zitzewitz (2007). “Partisan Impacts on the Economy: Evidence from Prediction Markets and Close Elections.” *Quarterly Journal of Economics* 122(2): 807–829.
- Yu, Y. (2017). “What Types of Financial Assets Do People Hold?” Federal Reserve Bank of St. Louis, *On the Economy* blog, October 19.
- Swanson, E. T., J. Wang, and J. Wu (2025). “Monetary Policy and the Fed Information Effect: Evidence from Prediction Markets.” Working paper, University of California, Irvine.
- The Vanguard Group (2023). “Personalized Inflation Hedging: A closer look at your true consumer price index.” Vanguard Research, February.
- Vissing-Jørgensen, A. (2002). “Limited Asset Market Participation and the Elasticity of Intertemporal Substitution.” *Journal of Political Economy* 110(4): 825–853.
- Wolfers, J., and E. Zitzewitz (2004). “Prediction Markets.” *Journal of Economic Perspectives* 18(2): 107–126.